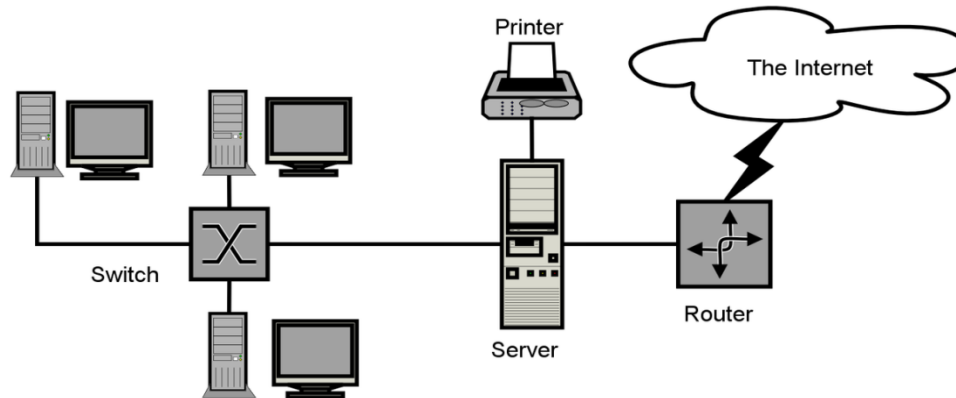


UNIT-1

Introduction to Networks

- ❖ A **computer network** is a group of **computers** that use a set of common communication protocols over digital interconnections for the purpose of sharing resources located on or provided by the **network devices**. They are identified by hostnames and **network** addresses.



State the need for networking:

Networks satisfy a broad range of purposes and meet various requirements.

- To provide sharing of geographically distant resources such as information, database etc.
- To provide sharing of physical resources printers, scanners and server systems ect.,
- To provide communication among users.
- To increase the reliability of processing capacity through backup and redundancy.
- To provide centralized management and allocation of resources.
- To provide distribution processing in which a task is divided among multiple computers.
- Faster problem solving.

Applications of Computer Network:

Network is used in different domains and different conditions. In that some of the major applications are listed below:

- Email programs (SMTP). They allow users to type messages at their local nodes and then send to someone on the network.
- File transfer protocol (FTP) This application facilities transfer of files from one computer to another e.g. from a client to a server.
- Terminal Emulation (TELNET) Use these actions to access and control operations on a remote machine.
- Groupware. etc.,

Define Data Communication:

“Data communication” is the exchange of data between two devices via some form of transmission medium such as a wire cable or wireless.

The communicating devices must be part of a communication system made up of a combination of hardware (physical equipment) and software (programs).

- The effectiveness of a data communications system depends on four fundamental characteristics:
 - Delivery
 - Accuracy
 - Timeliness
 - Jitter

1. Delivery: The system must deliver data to the correct destination. Data must be received by the intended device or user and only by that device or user.

2. Accuracy: The system must deliver the data accurately. Data that have been altered in transmission and left uncorrected are unusable.

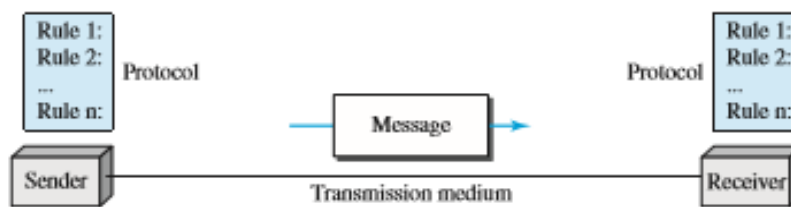
3. Timeliness: The system must deliver data in a timely manner. Data delivered late are useless. In the case of video and audio, timely delivery means delivering data as they are produced, in the same order that they are produced, and without significant delay. This kind of delivery is called real-time transmission.

4. Jitter: Jitter refers to the variation in the packet arrival time. It is the uneven delay in the delivery of audio or video packets. For example, let us assume that video packets are sent every 30-ms. If some of the packets arrive with 30-ms delay and others with 40-ms delay, an uneven quality in the video is the result.

Components of Data Communication:

❖ A data communications system has five components:

1. Message.
2. Sender.
3. Receiver.
4. Transmission medium.
5. Protocol.



1. Message: The message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.

2. Sender: The sender is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.

3. Receiver: The receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.

4. Transmission medium: The transmission medium is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.

5. Protocol: A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.

Data Representation:

❖ Data (information) today comes in different forms such as:

- Text
- Numbers
- Images
- Audio
- Video

- **Text:** In data communications, text is represented as a bit pattern, a sequence of bits (0s or 1s). Different sets of bit patterns have been designed to represent text symbols. Each set is called a code, and the process of representing symbols is called coding.
- **Numbers:** Numbers are also represented by bit patterns. However, a code such as ASCII is not used to represent numbers; the number is directly converted to a binary number to simplify mathematical operations.
- **Images:** Images are also represented by bit patterns. In its simplest form, an image is composed of a matrix of pixels (picture elements), where each pixel is a small dot. The size of the pixel depends on the resolution. For example, an image can be divided into 1000 pixels or 10,000 pixels.
- **Audio:** Audio refers to the recording or broadcasting of sound or music. Audio is by nature different from text, numbers, or images. It is continuous, not discrete. Even when we use a microphone to change voice or music to an electric signal, we create a continuous signal.
- **Video:** Video refers to the recording or broadcasting of a picture or movie. Video can either be produced as a continuous entity (e.g., by a TV camera), or it can be a combination of images, each a discrete entity, arranged to convey the idea of motion.

❖ Data Flow(Transmission modes):

Communication between two devices can be different ways:

- Simplex
- Half-duplex
- Full-duplex

- **Simplex:** In simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit; the other can only receive.

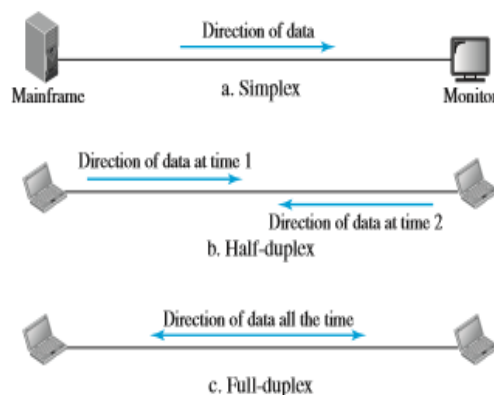
Example: Keyboards and traditional monitors are examples of simplex devices.

- **Half-Duplex:** In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa.

Example: Walkie-talkies and CB (citizens band) radios are both half-duplex systems.

- **Full-Duplex:** In full-duplex mode (also called duplex), both stations can transmit and receive simultaneously. The full-duplex mode is like a two-way street with traffic flowing in both directions at the same time. This sharing can occur in two ways: Either the link must contain two physically separate transmission paths, one for sending and the other for receiving; or the capacity of the channel is divided between signals traveling in both directions.

Example: Telephone network. When two people are communicating by a telephone line, both can talk and listen at the same time. The full-duplex mode is used when communication in both directions is required all the time. The capacity of the channel, however, must be divided between the two directions.



Define Network:

- A **network** is the interconnection of a set of devices capable of communication.
- In this definition, a device can be a **host** (or an *end system* as it is sometimes called) such as a large computer, desktop, laptop, workstation, cellular phone, or security system.
- A device in this definition can also be a **connecting device** such as a router, switch, modem and etc.,
- These devices in a network are connected using wired or wireless transmission media such as cable or air.

Network Criteria:

- A network must be able to meet a certain number of criteria.

The most important of these are:

- Performance
 - Reliability
 - Security
- **Performance:** Performance can be measured in many ways, including transit time and response time. Transit time is the amount of time required for a message to travel from one device to another. Response time is the elapsed time between an inquiry and a response. The performance of a network depends on a number of factors, including the number of users, the type of transmission medium, the capabilities of the connected hardware, and the efficiency of the software. Performance is often evaluated by two networking metrics: throughput and delay.
 - **Reliability:** In addition to accuracy of delivery, network reliability is measured by the frequency of failure, the time it takes a link to recover from a failure, and the network's robustness in a catastrophe.
 - **Security:** Network security issues include protecting data from unauthorized access, protecting data from damage and development, and implementing policies and procedures for recovery from breaches and data losses.

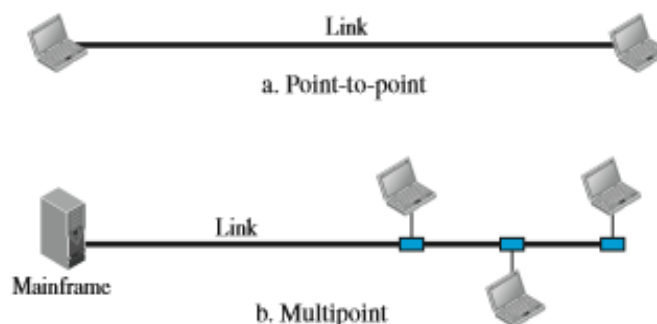
Physical Structures:

Before discussing networks, we need to define some network attributes.

Type of Connection:

A network is two or more devices connected through links. A link is a communications pathway that transfers data from one device to another. For visualization purposes, it is simplest to imagine any link as a line drawn between two points. For communication to occur, two devices must be connected in some way to the same link at the same time.

There are two possible types of connections: point-to-point and multipoint.



- **Point-to-Point:** A point-to-point connection provides a dedicated link between two devices. The entire capacity of the link is reserved for transmission between those two devices. Most point-to-point connections use an actual length of wire or cable to connect the two ends, but other options, such as microwave or satellite links, are also possible. When we change television channels by infrared remote control, we are establishing a point-to-point connection between the remote control and the television's control system.
- **Multipoint:** A multipoint (also called multidrop) connection is one in which more than two specific devices share a single link.

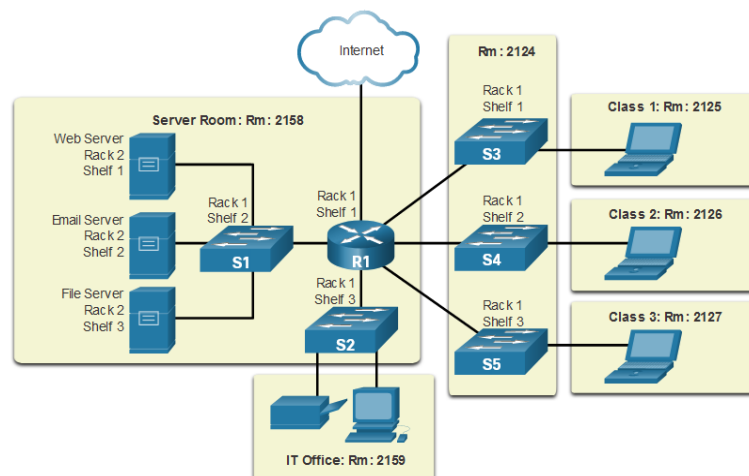
❖ Network Topology

Network topology is the layout of the connections formed between the nodes (end devices), simply network topology is also known as structure of a network.

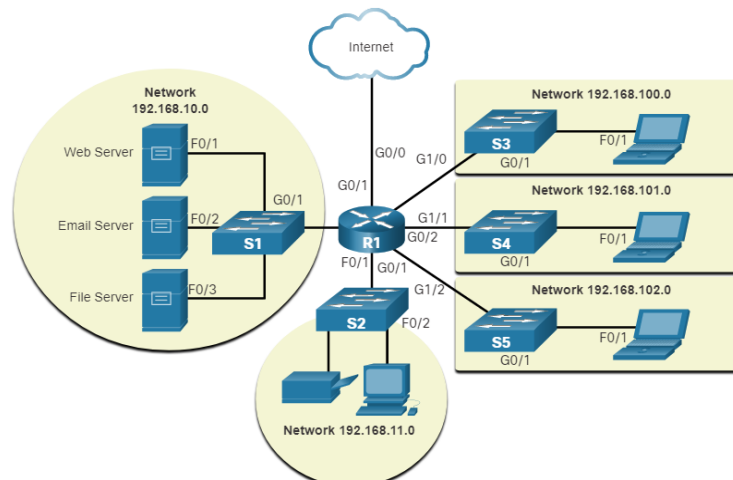
Two types of Topologies:

- Physical Topology
- Logical Topology

- **Physical topology:** Physical topology diagrams illustrate the physical location of nodes, intermediary devices and cable connections.

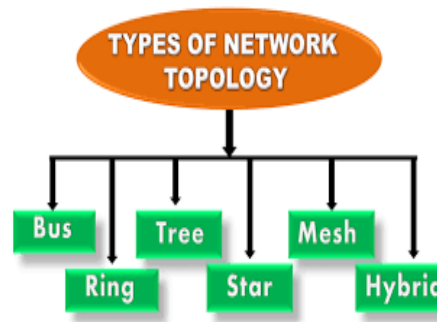


- **Logical topology:** Logical topology diagrams illustrate devices, ports, and the addressing scheme of the network.



Types of Physical Topology:

- ❖ Different categories of physical topology
 - Mesh topology
 - Star topology
 - Bus topology
 - Tree topology
 - Ring topology
 - Hybrid topology



❖ Mesh topology:

- In a mesh topology, every device has a dedicated point-to-point link to every other device.
- The term dedicated means that the link carries traffic only between the two devices it connects.
- To find the number of physical links in a fully connected mesh network with n nodes, we first consider that each node must be connected to every other node.
- Node 1 must be connected to $n - 1$ nodes, node 2 must be connected to $n - 1$ nodes, and finally node n must be connected to $n - 1$ nodes. We need $n(n - 1)$ physical links. However, if each physical link allows communication in both directions (duplex mode), we can divide the number of links by 2. In other words, we can say that in a mesh topology, we need $n(n - 1) / 2$ no. of links.

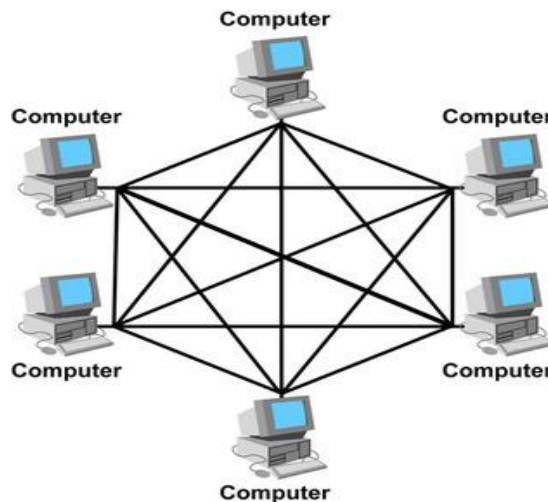


Fig: Mesh Topology

Advantages:

- Dedicated links (Direct communication) between each node.
- There is no traffic problem.
- It is robust.
- It is very secure.
- Fault identification easy.
- Adding nodes (devices) is easy.

Disadvantages:

- The amount of cabling and the number of I/O ports required.
- Cost expensive.
- Difficult to install and reconnecting.

❖ Star topology:

- In a star topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub.
- The devices are not directly linked to one another.
- Unlike a mesh topology, a star topology does not allow direct traffic between devices.
- The controller acts as an exchange: If one device wants to send data to another, it sends the data to the controller, which then relays the data to the other connected device.

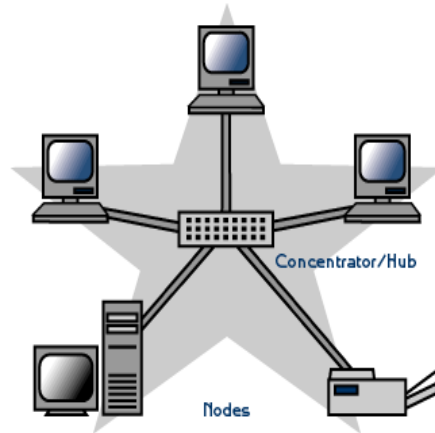


Fig: Star Topology

Advantages:

- It is less expensive than a mesh topology.
- It is easy to install and reconfigure.
- It is robust (If one link fails; only that link is affected. All other links remain active).
- Fault identification easy.

Disadvantages:

- A star topology is the dependency of the whole topology on one single point, the hub.
- If the hub goes down, the whole system is dead.
- It requires far less cable than a mesh; each node must be linked to a central hub. For this reason, often more cabling is required in a star than in some other topologies.

❖ Bus topology:

- A bus topology, on the other hand, is multipoint. One long cable acts as a backbone to link all the devices in a network.
- Nodes are connected to the bus cable by drop lines and taps.
- A drop line is a connection running between the device and the main cable.
- A tap is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core.
- As a signal travels along the backbone, some of its energy is transformed into heat. Therefore, it becomes weaker and weaker as it travels farther and farther. For this reason there is a limit on the number of taps a bus can support and on the distance between those taps.

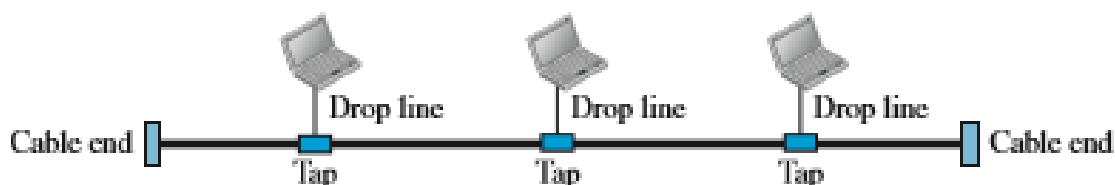


Fig: Bus Topology

Advantages:

- Ease of installation.
- Uses less cabling than mesh or star topologies.
- It is Cheap(Less cost).

Disadvantages:

- Difficult reconnection and fault isolation.
- Difficult to add new devices.
- A fault or break in the bus cable stops all transmission.

❖ Ring topology:

- In a ring topology, each device has a dedicated point-to-point connection with only the two devices on either side of it.
- A signal is passed along the ring in one direction, from device to device, until it reaches its destination.
- Each device in the ring incorporates a repeater. When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along

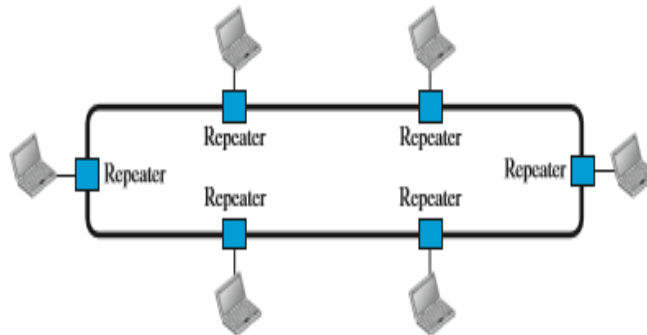


Fig: Ring Topology

Advantages:

- Easy to install and reconfigure.
- Fault identification easy.

Disadvantages:

- It is a unidirectional.
- If anyone device fails in the ring topology the entire network fails.
- Time delay (signal can travel long distance because each device in a ring can act as a repeater).

❖ Tree topology:

- Tree topology is the combination of the bus and the star topology.
- Tree topology allows users to have many servers on the network.

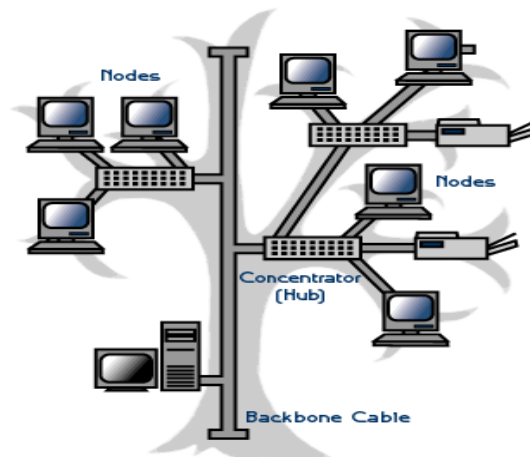


Fig: Tree Topology

- Tree topology follows a hierarchical pattern whereby each level is connected to the next higher level in a symmetrical pattern.
- It connects multiple star topologies to other star topology network.
- Tree topology is the best when the network is large and not for a small network because it is a waste of cables to use it.

Advantages:

- Point-to-point link for individual segments.
- Supported by several hardware and software vendors.
- Expansion of nodes is easy.
- Fault identification is also easy.
- Tree topology supports large networks.

Disadvantages:

- If the backbone line breaks, the entire segment goes down.
- More difficult to configure and wire than other topologies.
- Cost expensive.
- More cable is required.
- Difficult to maintain large networks.

❖ Hybrid topology:

- Hybrid topology is an integration of two or more different topologies to form a resultant topology.
- This combination of topologies is done according to the requirements of the organization.

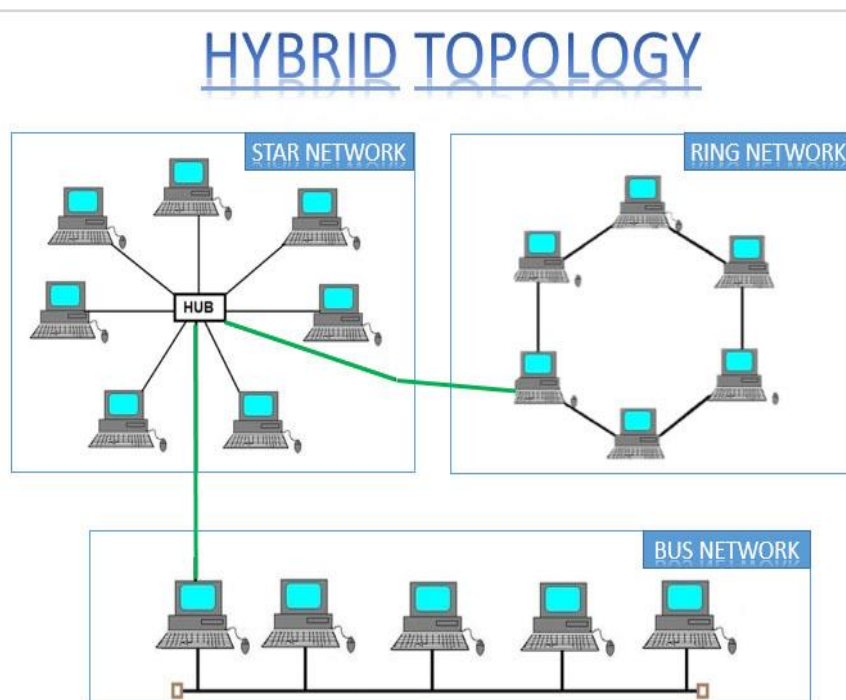


Fig: Hybrid Topology

Advantages:

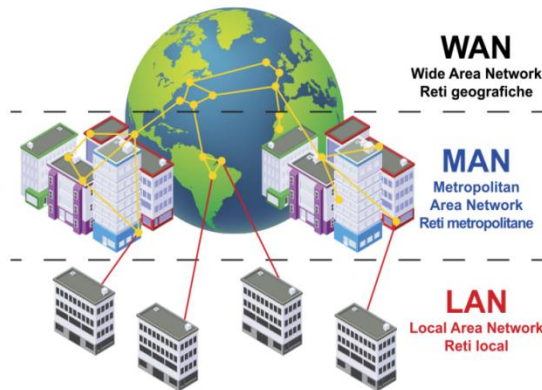
- It is reliable.
- Effective.
- Scalable.
- Flexible.

Disadvantages:

- Complex in design.
- Cost effective.

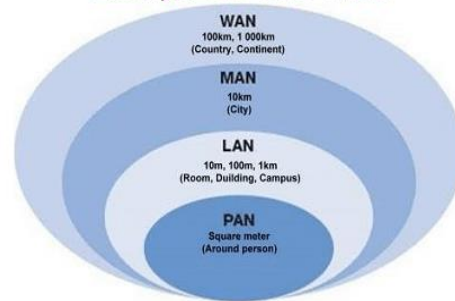
Network types:

- After defining networks in the previous section and discussing their physical structures, we need to discuss different types of networks we encounter in the world today.
- The criteria of distinguishing one type of network from another are difficult and sometimes confusing.
- We use a few criteria such as “size, geographical coverage, and ownership” to make this distinction.
- Networks are classified into different ways:
 - PAN(Personal Area Network)
 - LAN(Local Area Network)
 - MAN(Metropolitan Area Network)
 - WAN(Wide Area Network)



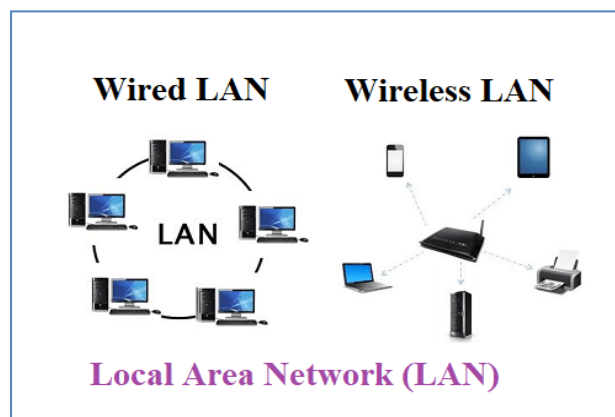
Network - Types of Computer Network

LAN, MAN and WAN



❖ LAN (Local Area Network):

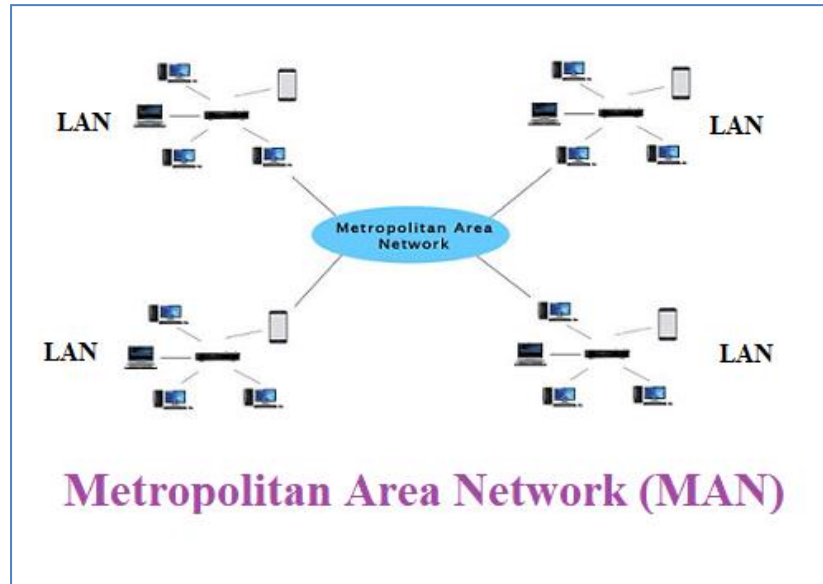
- Local area networks, generally called LANs, are privately-owned networks within a single building or campus of up to a few kilometers in size.
- They are widely used to connect personal computers and workstations in company offices and factories to share resources (e.g., printers) and exchange information.
- LANs are restricted in size, which means that the worst-case transmission time is bounded and known in advance.
- Traditional LANs run at speeds of 10 Mbps to 100 Mbps, have low delay (microseconds or nanoseconds), and make very few errors. Newer LANs operate at up to 10 Gbps.



❖ MAN (Metropolitan Area Network):

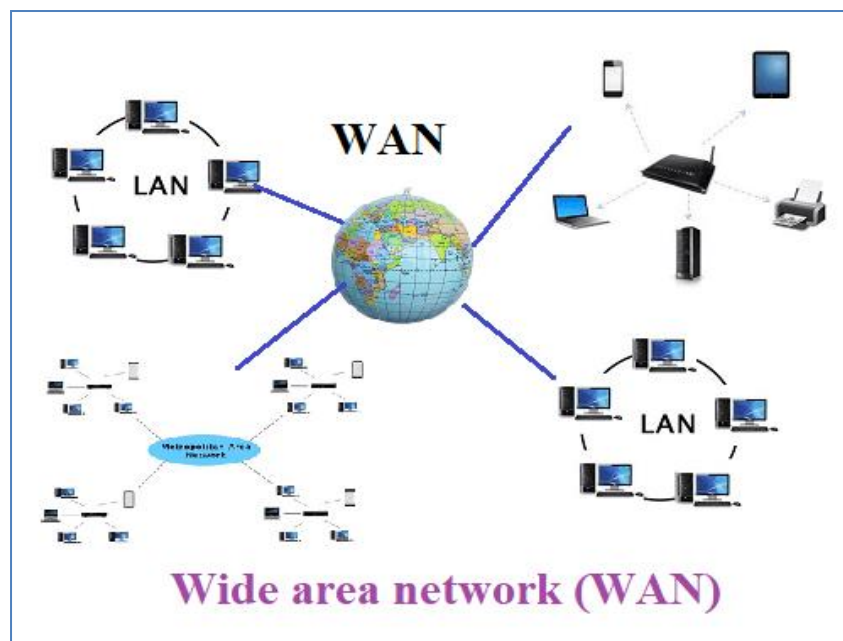
- A metropolitan area network, or MAN, covers a city.
- The best-known example of a MAN is the cable television network available in many cities.
- This system grew from earlier community antenna systems used in areas with poor over-the-air television reception.

- In these early systems, a large antenna was placed on top of a nearby hill and signal was then piped to the subscribers' houses.



❖ WAN(Wide Area Network):

- A wide area network, or WAN, spans a large geographical area, often a country or continent.
- It contains a collection of machines intended for running user (i.e., application) programs.
- The hosts are connected by a communication subnet, or just simply subnet.
- The hosts are owned by the customers (e.g., people's personal computers), whereas the communication subnet is typically owned and operated by a telephone company or Internet service provider.
- The job of the subnet is to carry messages from host to host, just as the telephone system carries words from speaker to listener.
- In most wide area networks, the subnet consists of two distinct components: transmission lines and switching elements.
- Transmission lines move bits between machines. They can be made of copper wire, optical fiber, or even radio links.
- Switching elements are specialized computers that connect three or more transmission lines.

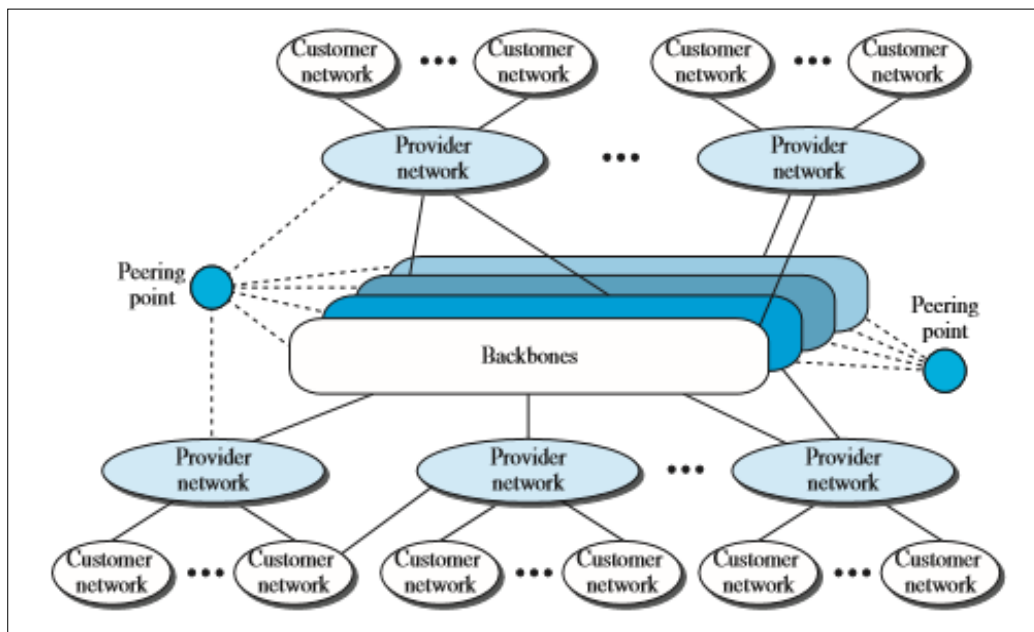


Differences Between LAN, MAN and WAN:

BASIS	LAN	MAN	WAN
Full Form	Local Area Network	Metropolitan Area Network	Wide Area Network
Range	Range is 1 to 10 km	Range is 10 to 100 km	Range is beyond 100 km
Media used	Uses guided media (wired)	Uses guided media(wired) as well as unguided media(wireless)	Uses unguided media (wireless)
Transmission Speed	High	Moderate	Low
Cost	Cheaper	Costly	More expensive
Equipment needed	NIC, Switch/Hub	Modem and Router	Microwave, Radio and Satellite etc.,
Design and maintenance	Easy	Difficult	Difficult
Ownership of Network	Private	Private or Public	Private or Public

The Internet:

- An internet (note the lowercase i) is two or more networks that can communicate with each other.
- The most notable internet is called the Internet (uppercase I), and is composed of thousands of interconnected networks. The following figure shows a conceptual (not geographical) view of the Internet.



- Internet as several backbones, provider networks (ISP's), and customer networks.

- At the top level, the backbones are large networks owned by some communication companies such as Sprint, Verizon (MCI), AT&T, and NTT. The backbone networks are connected through some complex switching systems, called peering points.
- At the second level, there are smaller networks, called provider networks that use the services of the backbones for a fee. The provider networks are connected to backbones and sometimes to other provider networks.
- The customer networks are networks at the edge of the Internet that actually use the services provided by the Internet. They pay fees to provider networks for receiving services.
- Backbones and provider networks are also called Internet Service Providers (ISPs). The backbones are often referred to as international ISPs; the provider networks are often referred to as national or regional ISPs.

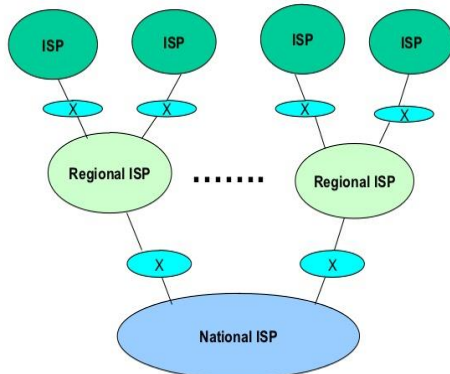


Fig 1. Structure of a national ISP

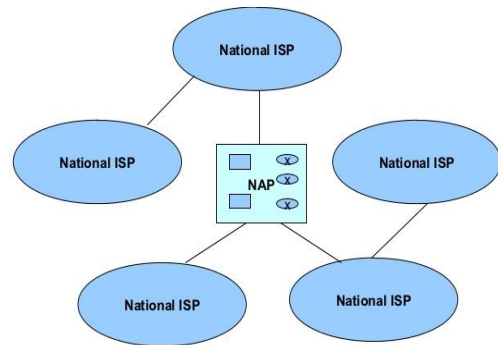
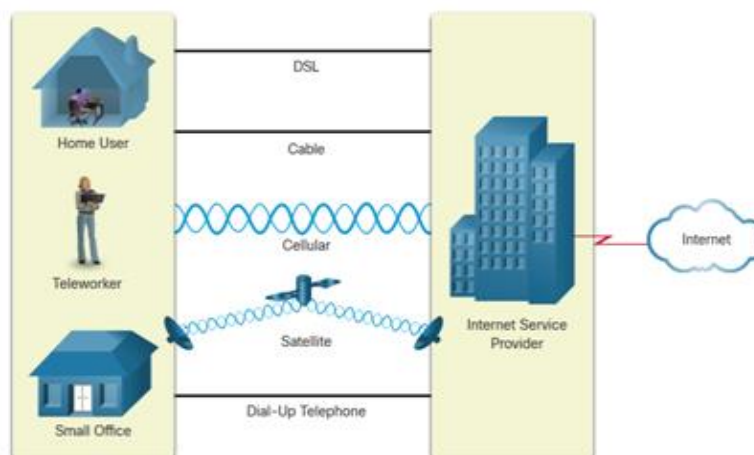


Fig.2. Interconnection of national ISPs

Internet access technologies:

- ❖ The Internet today is an internetwork that allows any user to become part of it.
- ❖ The user, however, needs to be physically connected to an ISP.
- ❖ There are many ways to connect users and organizations to the internet:
 - Using Telephone Networks
 - Dial-up service.
 - DSL Service.
 - Using Cable Networks.
 - Using Wireless Networks.
 - Direct Connection to the Internet.



❖ **Using Telephone Networks:**

Today most residences and small businesses have telephone service, which means they are connected to a telephone network. This can be done in two ways.

- **Dial-up service.** The first solution is to add to the telephone line a modem that converts data to voice. The software installed on the computer dials the ISP and imitates making a telephone connection. Unfortunately, the dial-up service is very slow, and when the line is used for Internet connection, it cannot be used for telephone (voice) connection. It is only useful for small residences.
- **DSL Service.** Since the advent of the Internet, some telephone companies have upgraded their telephone lines to provide higher speed Internet services to residences or small businesses. The DSL service also allows the line to be used simultaneously for voice and data communication.

❖ **Using Cable Networks :**

More and more residents over the last two decades have begun using cable TV services instead of antennas to receive TV broadcasting. The cable companies have been upgrading their cable networks and connecting to the Internet. A residence or a small business can be connected to the Internet by using this service. It provides a higher speed connection, but the speed varies depending on the number of neighbors that use the same cable.

❖ **Using Wireless Networks:**

Wireless connectivity has recently become increasingly popular. A household or a small business can use a combination of wireless and wired connections to access the Internet. With the growing wireless WAN access, a household or a small business can be connected to the Internet through a wireless WAN.

❖ **Direct Connection to the Internet :**

A large organization or a large corporation can itself become a local ISP and be connected to the Internet. This can be done if the organization or the corporation leases a high-speed WAN from a carrier provider and connects itself to a regional ISP. For example, a large university with several campuses can create an internetwork and then connect the internetwork to the Internet.

Internet History:

- All Public and Private Individuals as well as various organizations such as government agencies, schools, research facilities, corporations, and libraries in more than 100 countries use the Internet. Millions of people are users. Yet this extraordinary communication system only came into being in 1969.
- In the mid-1960s, mainframe computers in research organizations were standalone devices. Computers from different manufacturers were unable to communicate with one another. The Advanced Research Projects Agency (ARPA) in the Department of Defense (DoD) was interested in finding a way to connect computers so that the researchers they funded could share their findings, thereby reducing costs and eliminating duplication of effort.
- In 1967, at an Association for Computing Machinery (ACM) meeting, ARPA presented its ideas for ARPANET, a small network of connected computers. The idea was that each host computer (not necessarily from the same manufacturer) would be attached to a specialized computer, called an interface message processor (IMP). The IMPs, in turn, would be connected to one another. Each IMP had to be able to communicate with other IMPs as well as with its own attached host.
- By 1969, ARPANET was a reality. Four nodes, at the University of California at Los Angeles (UCLA), the University of California at Santa Barbara (UCSB), Stanford Research Institute (SRI), and the University of

Utah, were connected via the IMPs to form a network. Software called the Network Control Protocol (NCP) provided communication between the hosts.

- In 1972, Vint Cerf and Bob Kahn, both of whom were part of the core ARPANET group, collaborated on what they called the Internetting Project. Cerf and Kahn's landmark 1973 paper outlined the protocols to achieve end-to-end delivery of packets. This paper on Transmission Control Protocol (TCP) included concepts such as encapsulation, the datagram, and the functions of a gateway.
- Shortly thereafter, authorities made a decision to split TCP into two protocols: Transmission Control Protocol (TCP) and Internetworking Protocol (IP). IP would handle datagram routing while TCP would be responsible for higher-level functions such as segmentation, reassembly, and error detection. The internetworking protocol became known as TCPIIP.

The Internet Today:

- The Internet has come a long way since the 1960s.
- The Internet today is not a simple hierarchical structure.
- It is made up of many wide- and local-area networks joined by connecting devices and switching stations.
- It is difficult to give an accurate representation of the Internet because it is continually changing-new networks are being added, existing networks are adding addresses, and networks of defunct companies are being removed.
- Today most end users who want Internet connection use the services of Internet service providers (ISPs).

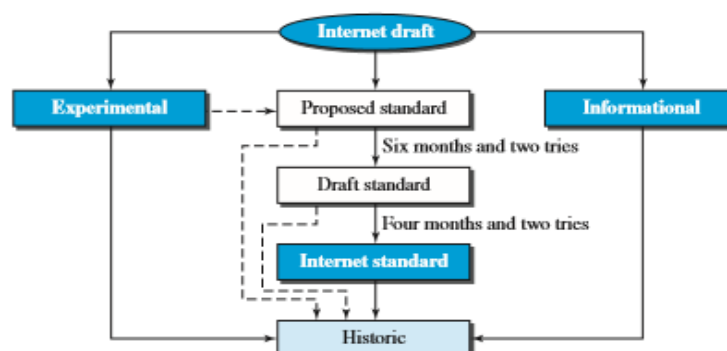
❖ INTERNET STANDARDS AND ADMINISTRATION:

Internet Standards:

- An **Internet standard** is a thoroughly tested specification that is useful to and adhered to by those who work with the Internet.
- It is a formalized regulation that must be followed. There is a strict procedure by which a specification attains Internet standard status.
- A specification begins as an **Internet draft**. An Internet draft is a working document (a work in progress).
- Upon recommendation from the Internet authorities, a draft may be published as a **Request for Comment (RFC)**.
- RFCs go through maturity levels and are categorized according to their requirement level.

➤ Maturity Levels

➤ Requirement Levels



❖ Maturity Levels:

An RFC, during its lifetime, falls into one of six maturity levels: proposed standard, draft standard, Internet standard, historic, experimental, and informational.

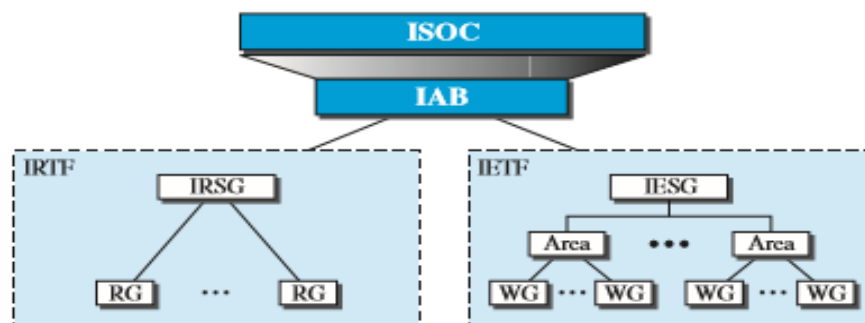
- **Proposed Standard.** A proposed standard is a specification that is stable, well understood, and of sufficient interest to the Internet community. At this level, the specification is usually tested and implemented by several different groups.
- **Draft Standard.** A proposed standard is elevated to draft standard status after at least two successful independent and interoperable implementations. Barring difficulties, a draft standard, with modifications if specific problems are encountered, normally becomes an Internet standard.
- **Internet Standard.** A draft standard reaches Internet standard status after demonstrations of successful implementation.
- **Historic.** The historic RFCs are significant from a historical perspective. They either have been superseded by later specifications or have never passed the necessary maturity levels to become an Internet standard.
- **Experimental.** An RFC classified as experimental describes work related to an experimental situation that does not affect the operation of the Internet. Such an RFC should not be implemented in any functional Internet service.
- **Informational.** An RFC classified as informational contains general, historical, or tutorial information related to the Internet. It is usually written by someone in a non-Internet organization, such as a vendor.

❖ Requirement Levels

RFCs are classified into five requirement levels: required, recommended, elective, limited use, and not recommended.

- **Required.** An RFC is labeled required if it must be implemented by all Internet systems to achieve minimum conformance. For example, IP and ICMP are required protocols.
- **Recommended.** An RFC labeled recommended is not required for minimum conformance; it is recommended because of its usefulness. For example, FTP and TELNET are recommended protocols.
- **Elective.** An RFC labeled elective is not required and not recommended. However, a system can use it for its own benefit.
- **Limited Use.** An RFC labeled limited use should be used only in limited situations. Most of the experimental RFCs fall under this category.
- **Not Recommended.** An RFC labeled not recommended is inappropriate for general use. Normally a historic (deprecated) RFC may fall under this category.

❖ Internet Administration



- The Internet, with its roots primarily in the research domain, has evolved and gained a broader user base with significant commercial activity.
- Various groups that coordinate Internet issues have guided this growth and development.
- The following diagram shows the general organization of Internet administration.
- **The Internet Society (ISOC)** accomplishes this through maintaining and supporting other Internet administrative bodies such as IAB, IETF, IRTF, and IANA. ISOC also promotes research and other scholarly activities relating to the Internet.
- **The Internet Architecture Board (IAB)** is the technical advisor to the ISOC. The main purposes of the IAB are to oversee the continuing development of the TCP/IP Protocol Suite and to serve in a technical advisory capacity to research members of the Internet community.
- **The Internet Engineering Task Force (IETF)** is a forum of working groups managed by the Internet Engineering Steering Group (IESG). IETF is responsible for identifying operational problems and proposing solutions to these problems.
- **The Internet Research Task Force (IRTF)** is a forum of working groups managed by the Internet Research Steering Group (IRSG). IRTF focuses on long-term research topics related to Internet protocols, applications, architecture, and technology.

❖ **Protocols and Standards**

- In computer networks, communication occurs between entities in different systems. An entity is anything capable of sending or receiving information. For communication to occur, the entities must agree on a protocol.
- A protocol is a set of rules that govern data communications. A protocol defines what is communicated, how it is communicated, and when it is communicated.
- The key elements of a protocol are :
- **Syntax (What is communicated):** The term syntax refers to the structure or format of the data, meaning the order in which they are presented.
- **Semantics (How it is communicated):** The word semantics refers to the meaning of each section of bits.
- **Timing (When it is communicated):** The term timing refers to two characteristics: when data should be sent and how fast they can be sent.

❖ **Standards:**

- Standards provide guidelines to manufacturers, vendors, government agencies, and other service providers to ensure the kind of interconnectivity necessary in today's marketplace and in international communications.
- Standards are developed through the cooperation of standards creation committees, forums, and government regulatory agencies.

❖ **Standards Creation Committees**

- International Organization for Standardization (ISO).
- International Telecommunication Union-Telecommunication Standards Sector (ITU-T).
- American National Standards Institute (ANSI).
- Institute of Electrical and Electronics Engineers (IEEE).
- Electronic Industries Association (EIA).

❖ Forums:

- The forums work with universities and users to test, evaluate, and standardize new technologies. By concentrating their efforts on a particular technology, the forums are able to speed acceptance and use of those technologies in the telecommunications community. The forums present their conclusions to the standards bodies.

❖ Regulatory Agencies:

- All communications technology is subject to regulation by government agencies such as the Federal Communications Commission (FCC) in the United States.
- The purpose of these agencies is to protect the public interest by regulating radio, television, and wire/cable communications. The FCC has authority over interstate and international commerce as it relates to communications.

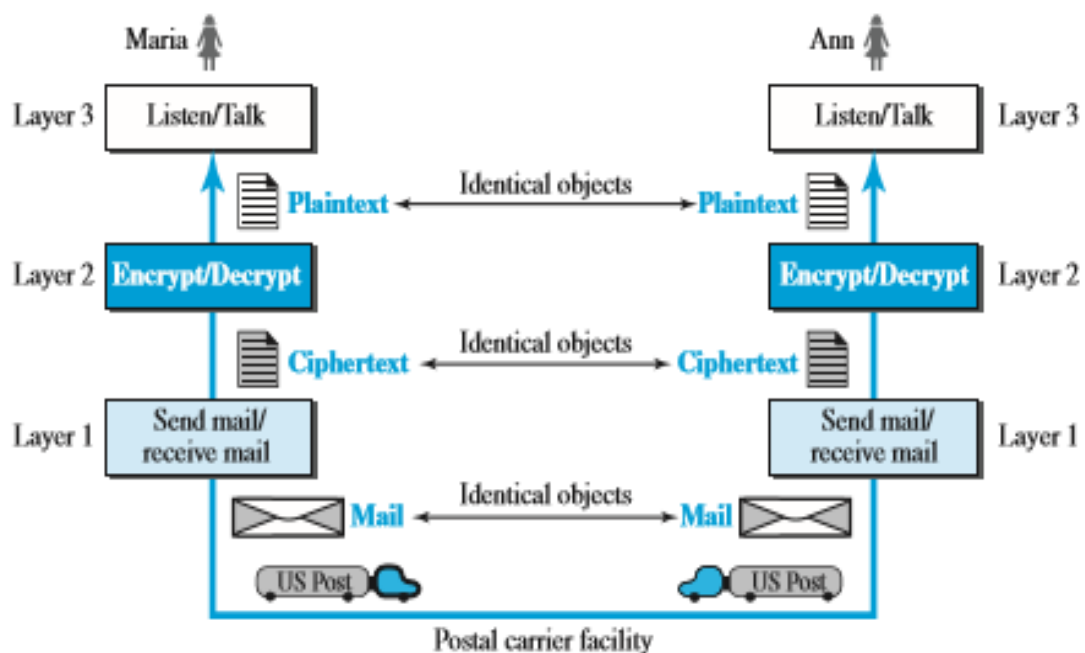
❖ PROTOCOL LAYERING

- In networking **“protocol”** defines the rules that both the sender and receiver and all intermediate devices need to follow to be able to communicate effectively.
 - When communication is simple, we may need only one simple protocol.
 - When the communication is complex, we may need to divide the task between different layers, in which case we need a protocol at each layer, or protocol layering.
- ❖ Let us develop two simple **“scenarios”** to better understand the need for protocol layering.

1. First Scenario (A single-layer protocol):



2. Second Scenario (A three-layer protocol):

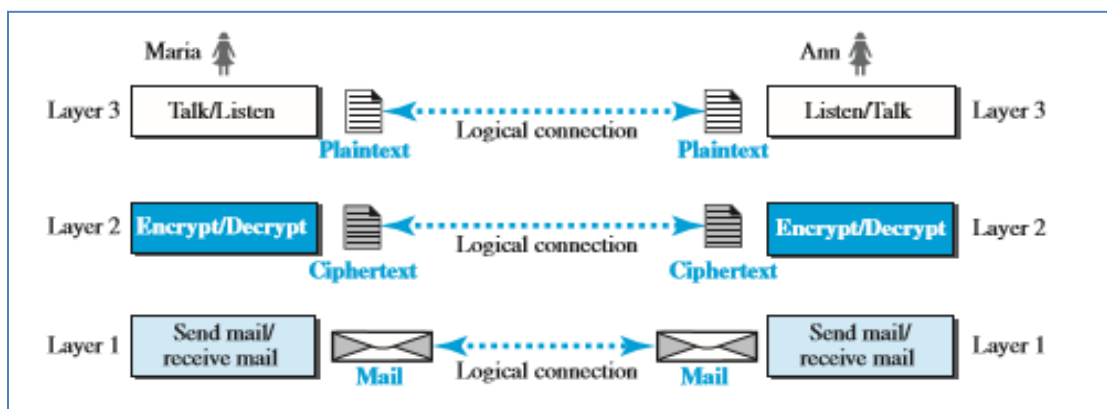


❖ Principles of Protocol Layering:

- **First Principle:** The first principle dictates that if we want bidirectional communication, we need to make each layer so that it is able to perform two opposite tasks, one in each direction. For example, the third layer task is to listen (in one direction) and talk (in the other direction). The second layer needs to be able to encrypt and decrypt. The first layer needs to send and receive mail.
- **Second Principle:** The second principle that we need to follow in protocol layering is that the two objects under each layer at both sites should be identical. For example, the object under layer 3 at both sites should be a plaintext letter. The object under layer 2 at both sites should be a cipher text letter. The object under layer 1 at both sites should be a piece of mail.

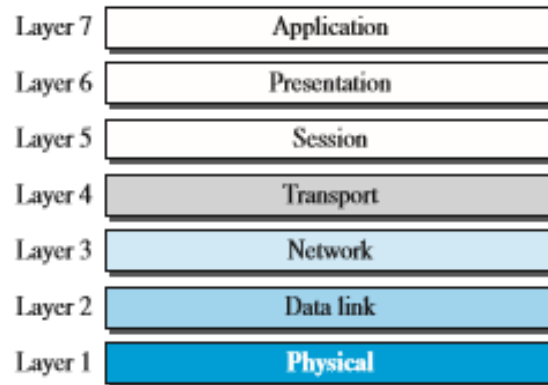
❖ Logical Connections:

- After following the above two principles, we can think about logical connection between each layer. This means that we have layer-to-layer communication.
- We will see that the concept of logical connection will help us better understand the task of layering we encounter in data communication and networking.

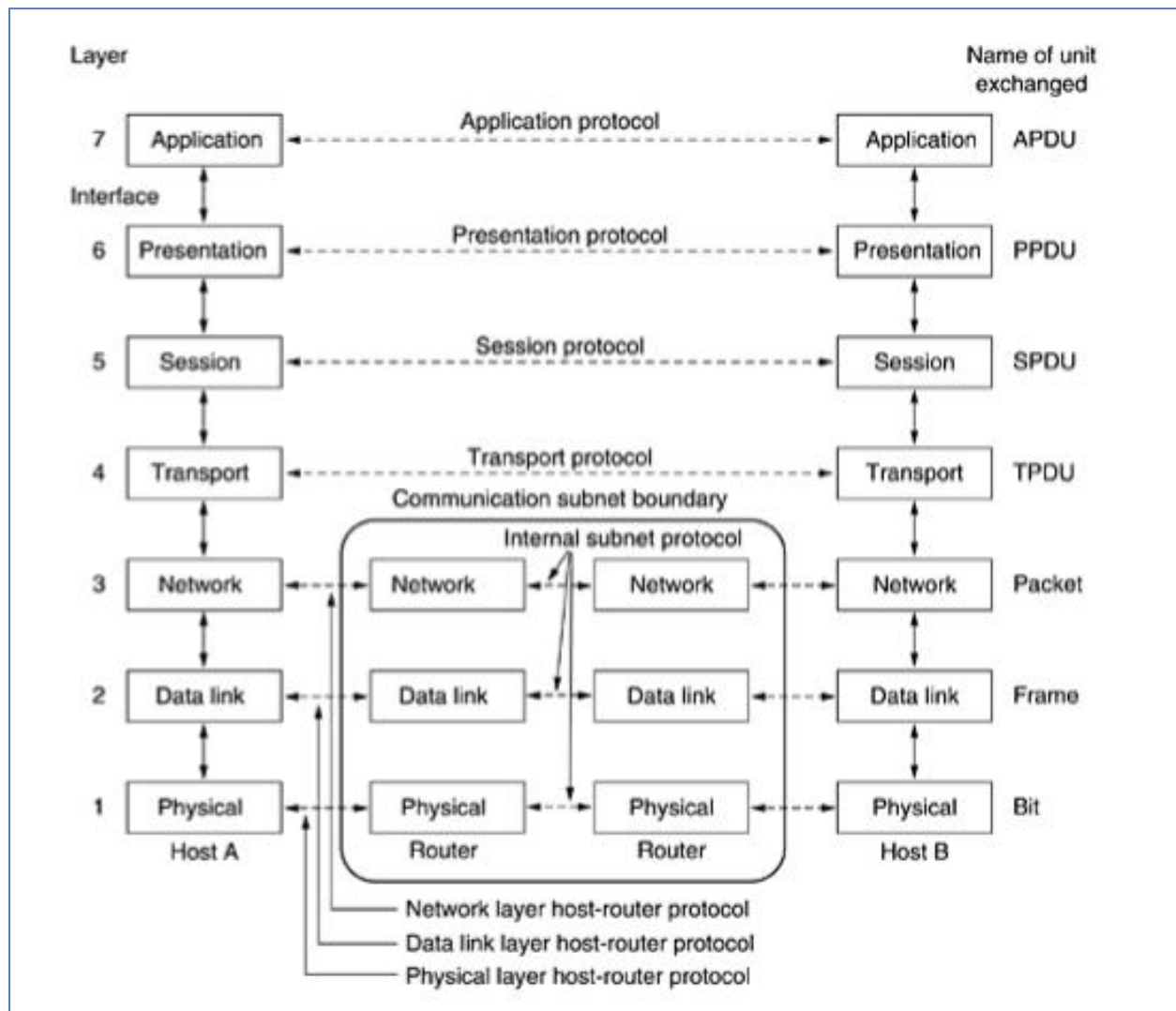


❖ THE OSI REFERENCE MODEL:

- An **ISO (International Organization for Standardization)** standard that covers all aspects of network communications is the **Open Systems Interconnection (OSI)** model. It was first introduced in the late 1970s.
- An open system is a set of protocols that allows any two different systems to communicate regardless of their underlying architecture.
- The purpose of the OSI model is to show how to facilitate communication between different systems without requiring changes to the logic of the underlying hardware and software.
- The OSI model is not a protocol; it is a model for understanding and designing a network architecture that is flexible, robust, and interoperable.
- The OSI model is a layered framework for the design of network systems that allows communication between all types of computer systems.
- It consists of seven separate but related layers, each of which defines a part of the process of moving information across a network.



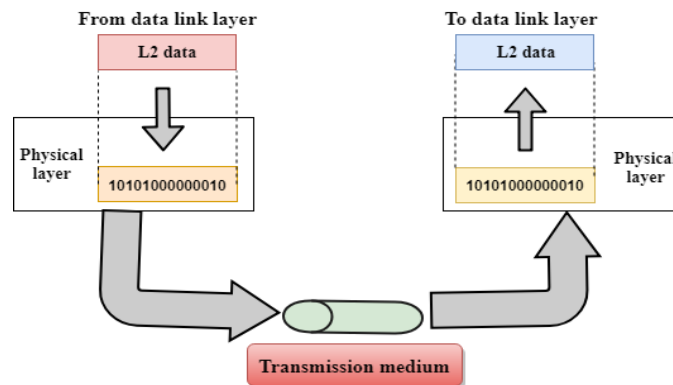
THE OSI Reference Model Architecture:



1. Physical Layer:

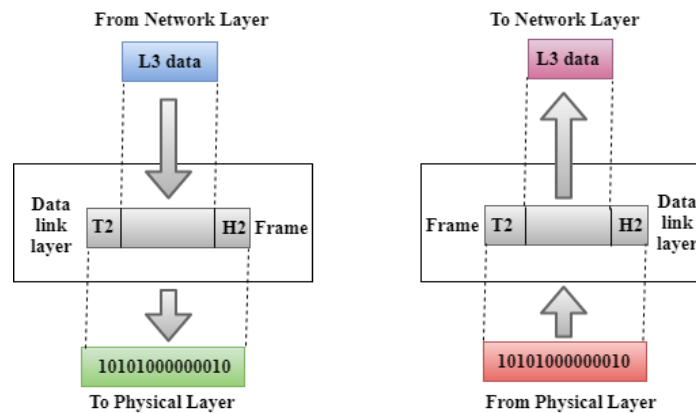
- The physical layer is responsible for transmitting raw bits from one node to another node over a communication channel or transmission medium.
- It converts frames from the data link layer into bits and at the receiving end, bits from the physical layer is converted into frames.
- For sending raw bits from source to destination, the source and destination nodes have to agree on a number of factors such as:

- What voltage constitutes a bit value '0' and what voltage constitutes a bit value '1'.
- What is the bit interval (one side sends a 1 bit; it is received by the other side as a 1 bit, not as a 0 bit).
- Whether the communication is in only one direction or both directions simultaneously.
- It also deals with the electrical specifications of the cables, connectors and interfaces.

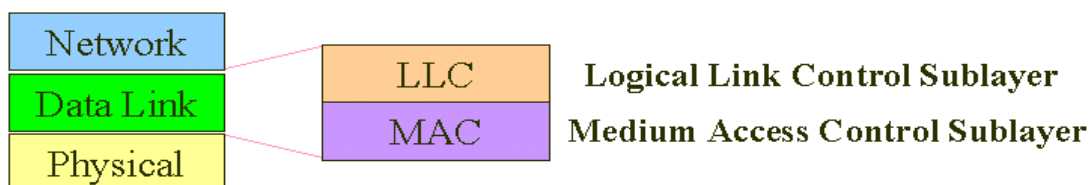


2. Data link Layer:

- The data link layer converts data packets from network layer into frames and at the receiving end, this layer converts frames into packets.
- In the data link layer manageable data units are called frames.
- The data link layer receives the data to be sent from the network layer, adds header and trailer to it which is now known as a frame.
- This frame is then transmitted to the physical layer.
- Switches are the layer 2 devices; they are specifying the unique identification of each device that resides on a local network.

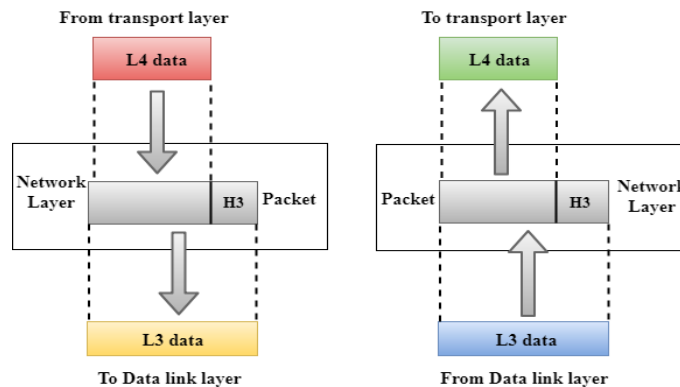


- Data link layer has two sub-layers:
 - **Logical Link Control:** It deals with protocols, flow-control, and error control
 - **Media Access Control:** It deals with actual control of media



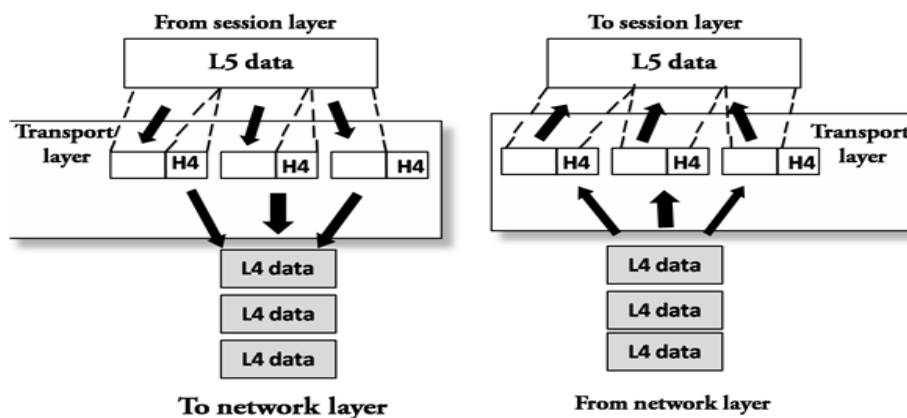
3. Network Layer:

- The network layer converts data segments into packets and at the receiving end, the packets are converted into data segments.
- This layer determines best paths (routing) for transmitting data from source to destination.
- It manages device addressing, tracks the location of devices on the network.
- Routers are the layer 3 devices; they are specified in this layer and used to provide the routing services within an internetwork.
- It provides flow-control functions across the computer-network interface (internetworking).



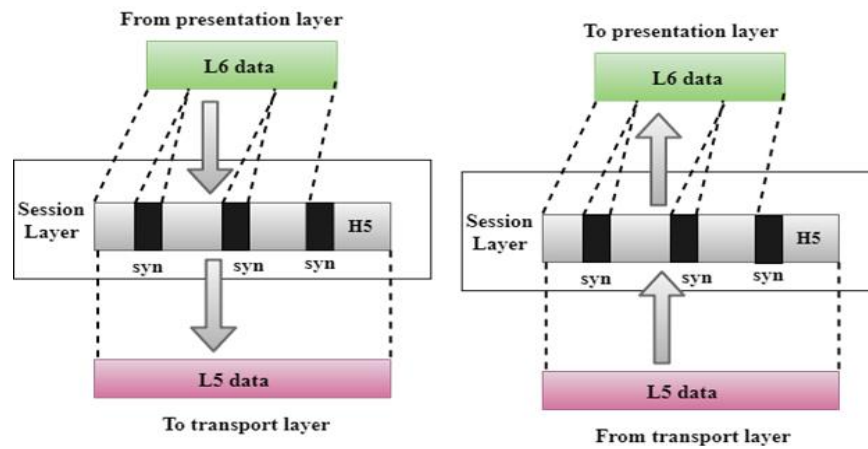
4. Transport layer:

- This layer converts data into smaller “segments” and at the receiving end; the segments can be converted into original data.
- The transport layer is responsible for source to destination (end-to-end) delivery of the entire message.
- It is also responsible for:
 - Retransmission of the data.
 - Error control.
 - Flow control.



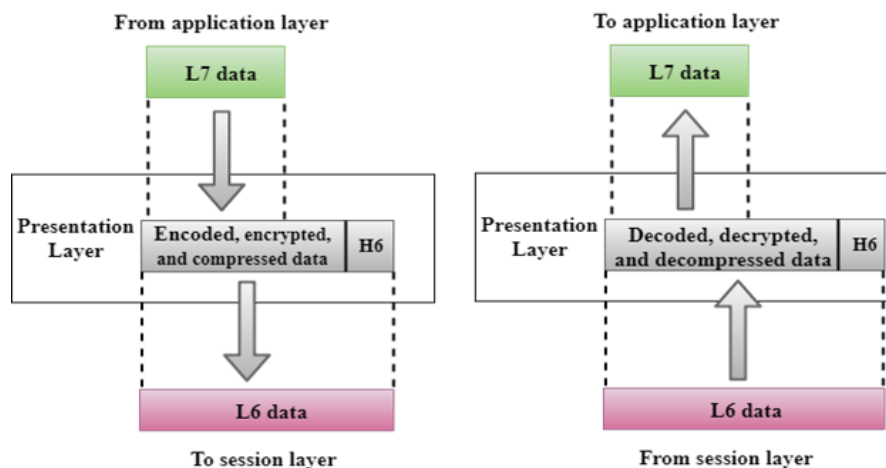
5. Session Layer:

- The Session layer is used to establish, maintain and synchronizes the interaction between communicating devices.
- Session layer acts as a dialog controller that creates a dialog between two processes or we can say that it allows the communication between two processes which can be either half-duplex or full-duplex.
- Session layer adds some checkpoints when transmitting the data in a sequence. If some error occurs in the middle of the transmission of data, then the transmission will take place again from the checkpoint. This process is known as Synchronization and recovery.



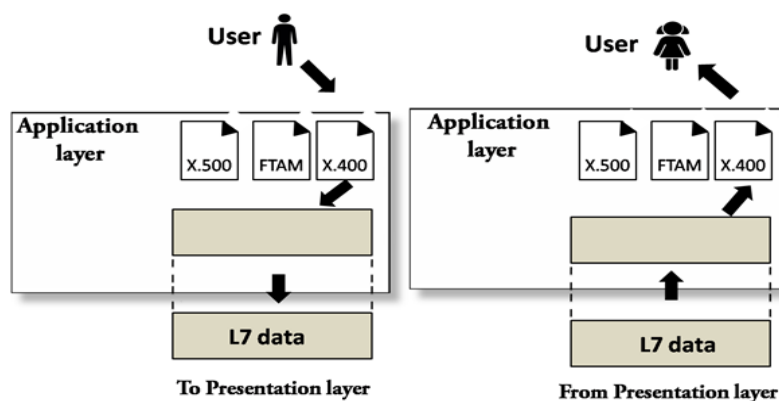
6. Presentation Layer:

- A Presentation layer is mainly concerned with the syntax and semantics of the information exchanged between the two systems.
- It translates the Application into network format and vice versa.
- It provides format, compression and encrypt/decrypt data to be send across a network.



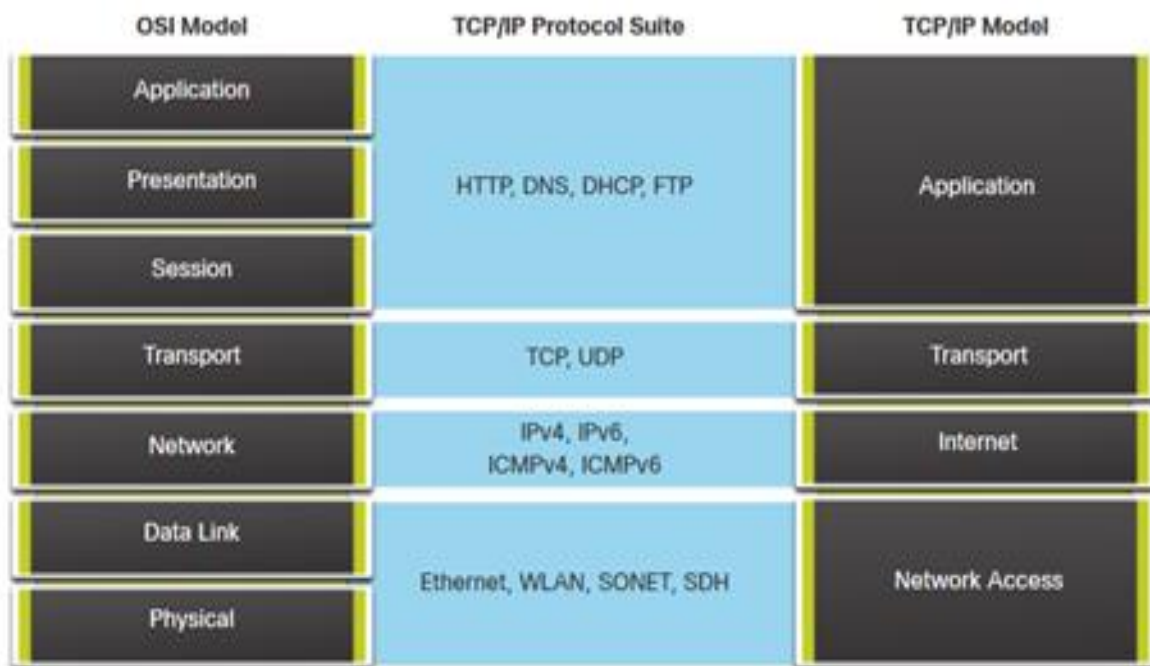
7. Application Layer:

- The application layer enables the user, either human or software access the network.
- It provides end user for processing of data and supports for services such as e-mail, file transfer, shared data base management, network software services and other types of distributed information services.
- This layer acts as an interface between end user and network.
- This layer provides the network services to the end-users.



❖ TCP/IP PROTOCOL SUITE (TCP/IP REFERENCE MODEL):

- The ARPANET was a research network sponsored by the DoD (U.S. Department of Defense). It eventually connected hundreds of universities and government installations, using leased telephone lines.
- When satellite and radio networks were added later, the existing protocols had trouble interworking with them, so new reference architecture was needed.
- Thus, the ability to connect multiple networks in a seamless way was one of the major design goals from the very beginning.
- This architecture later became known as the TCP/IP Reference Model, after its two primary protocols. It was first defined in (Cerf and Kahn, 1974).
- The TCP/IP (Transmission Control Protocol/Internet Protocol) is a protocol suite (a set of protocols organized in different layers) used in the Internet today.
- It is a hierarchical protocol made up of interactive modules, each of which provides a specific functionality.
- The term hierarchical means that each upper level protocol is supported by the services provided by one or more lower level protocols.
- The original TCP/IP protocol suite was defined as four software layers built upon the hardware. Today, however, TCP/IP is thought of as a five-layer model.



1. Network Access Layer:

- A network access layer is the lowest layer of the TCP/IP model.
- **“A network access layer is the combination of the Physical layer and Data Link layer defined in the OSI reference model”.**
- It defines how the data should be sent physically through the network.
- This layer is mainly responsible for the transmission of the data between two devices on the same network.
- The functions carried out by this layer are encapsulating the IP datagram into frames transmitted by the network and mapping of IP addresses into physical addresses.
- The protocols used by this layer are Ethernet, token ring, FDDI, X.25, frame relay.

2. Internet Layer:

- An internet layer is also known as the network layer.
- At network layer, the main protocol defined by TCP/IP is Internet Protocol (IP).
- Internet layer concerned with routing of data.
- The job of the internet layer is to deliver IP packets to the destination.
- Following are the protocols used in this layer are:
 - **IP Protocol:** IP protocol is used in this layer, and it is the most significant part of the entire TCP/IP suite.
 - **Address Resolution Protocol:** ARP is a network layer protocol which is used to find the physical address from the IP address.
 - **ICMP (Internet Control Message Protocol):** It is a mechanism used by the hosts or routers to send notifications regarding datagram problems back to the sender.

3. Transport Layer:

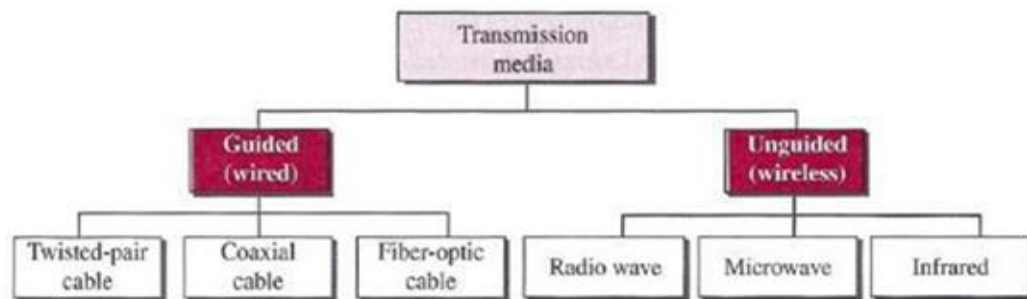
- The transport layer is responsible for the reliability (end-to-end data transmission), flow control, and error-correction of data which is being sent over the network.
- The two protocols used in the transport layer are :
 - Transmission control protocol(TCP)
 - User Datagram protocol (UDP).
- ❖ **Transmission control protocol(TCP):**
 - ☐ TCP is connection-oriented protocol.
 - ☐ TCP is a reliable.
 - ☐ TCP is identifying the loss of data and retransmit that data.
 - ☐ TCP will generate the acknowledgment that the data is received at the destination.
 - ☐ TCP also provides flow control and error control.
 - ☐ TCP is slower than UDP.
- ❖ **User Datagram protocol (UDP)**
 - ☐ UDP is a connectionless protocol.
 - ☐ UDP is not reliable.
 - ☐ Any segments that are lost are not resent.
 - ☐ There is no session establishment.
 - ☐ No acknowledgment that the data is received at the destination.
 - ☐ UDP is faster than TCP.

4. Application Layer:

- Application layer in the TCP/IP reference model is equal to top three layers in OSI model.
- The TCP/IP model does not have session and presentation layer.
- Communication at the application layer is between two processes.
- Process-to-process communication is the duty of application later.
- The application layer in the internet includes many predefined protocols, but a user can also create a pair of processes to be run at the two hosts.
- There are different application layer protocols are: HTTP,SMTP,FTP,TELNET,SSH,SNMP and DNS etc.,

❖ Transmission Media:

- The transmission medium can be defined as a pathway that can transmit information from a sender to a receiver. Transmission media are located below the physical layer and are controlled by the physical layer. Transmission media are also called communication channels.

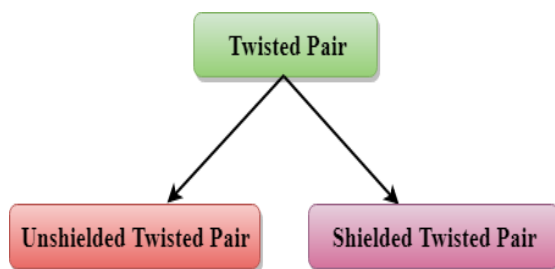


Guided Media (Wired):

- It is defined as the physical medium (Wired connection) through which the signals are transmitted. It is also known as Bounded media.

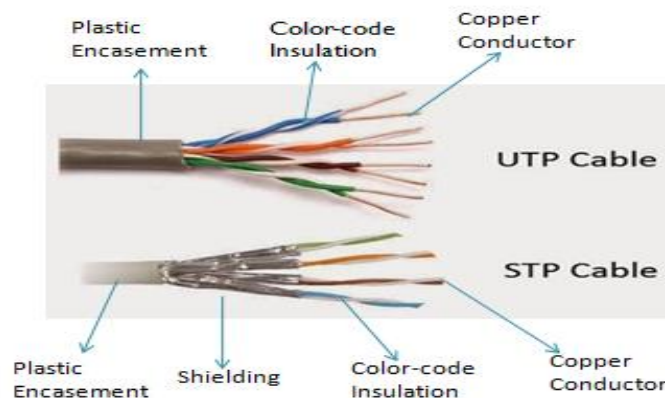
❖ Twisted pair:

- Twisted pair is a physical media made up of a pair of cables twisted with each other.
- Types of Twisted pair cables:



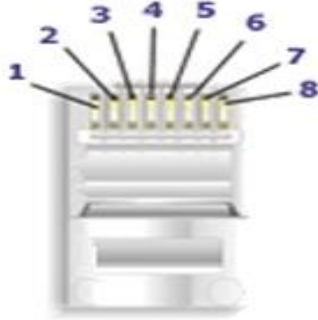
UTP Cable:

- Unshielded twisted pair (UTP) cables are found in many Ethernet networks and telephone systems.
- UTP is also the most common cable used in computer networking.
- It uses electric signal for transmission.
- This cable can carry both voice as well as data.
- Electronic Industries Association (EIA) has developed standards for UTP cable.

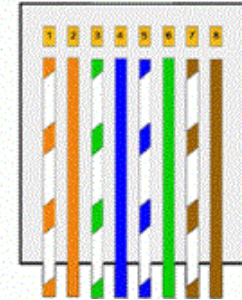


Making Straight UTP Cable :(TIA/EIA 568A wiring both sides)

Orange White on no 1
Orange on no 2
Green White on no 3
Blue on no 4
Blue White on no 5
Green on no 6
White Brown on no 7
Brown on no 8



STRAIGHT CABLE



Creating Cross UTP Cable: (TIA/EIA 568A is one side and TIA/EIA 568B is another side)

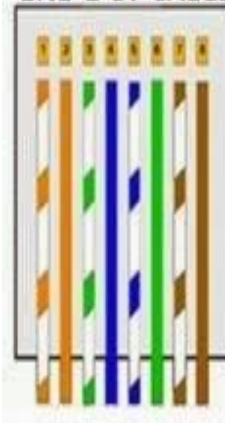
The first ends is same with straight cable :

- Orange White on no. 1
- Orange on no. 2
- Green White on no. 3
- Blue on no. 4
- Blue White on no. 5
- Green on no. 6
- White chocolate on no.7
- Brown on no. 8

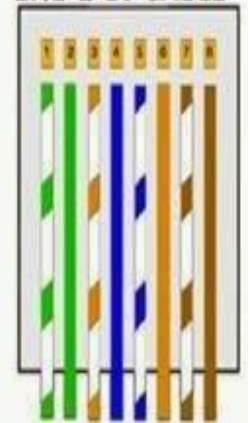
For the second end of the cable, the color composition is different from the first. the color arrangement is as follows :

- Green White on no. 1
- Green on no. 2
- Orange White on no. 3
- Blue on no. 4
- Blue White on no. 5
- Orange on no. 6
- White chocolate on no. 7
- Brown on no. 8

END 1 OF CABLE



END 2 OF CABLE



UTP Connectors:

The most common UTP connector is RJ45 (RJ stands for registered jack).The RJ45 is a keyed connector, meaning the connector can be inserted in only one way.



UTP Categories - Copper Cable

UTP Category	Data Rate	Max. Length	Cable Type	Application
CAT1	Up to 1Mbps	-	Twisted Pair	Old Telephone Cable
CAT2	Up to 4Mbps	-	Twisted Pair	Token Ring Networks
CAT3	Up to 10Mbps	100m	Twisted Pair	Token Rink & 10BASE-T Ethernet
CAT4	Up to 16Mbps	100m	Twisted Pair	Token Ring Networks
CAT5	Up to 100Mbps	100m	Twisted Pair	Ethernet, FastEthernet, Token Ring
CAT5e	Up to 1 Gbps	100m	Twisted Pair	Ethernet, FastEthernet, Gigabit Ethernet
CAT6	Up to 10Gbps	100m	Twisted Pair	GigabitEthernet, 10G Ethernet (55 meters)
CAT6a	Up to 10Gbps	100m	Twisted Pair	GigabitEthernet, 10G Ethernet (55 meters)
CAT7	Up to 10Gbps	100m	Twisted Pair	GigabitEthernet, 10G Ethernet (100 meters)

❖ **Advantages:**

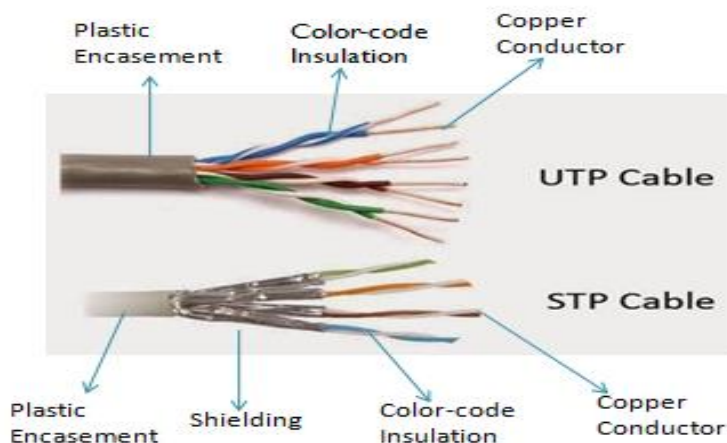
- Easy to install.
- Less cost expensive (Cheap).
- More flexible.
- Easy to use.

❖ **Disadvantages:**

- Cross talk is present.
- More noise produced.
- Supports short distance.

STP Cable:

- **STP Cabling** is twisted-pair **cabling** with additional shielding to reduce crosstalk and other forms of electromagnetic interference (EMI).
- The outer insulating jacket contains an inner braided copper mesh to shield the pairs of twisted **cables**, which themselves are wrapped in foil.



❖ **Advantages:**

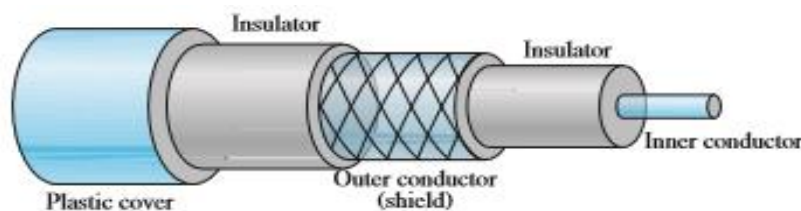
- Easy to install.
- Cross talk is eliminated.
- Less flexible.
- Easy to use.

❖ **Disadvantages:**

- Cost is high (expensive).
- Bulkier.

Co-axial cable:

- The outer metallic wrapping serves both as a shield against noise and as the second conductor, which completes the circuit.
- This outer conductor is also enclosed in an insulating sheath, and the whole cable is protected by a plastic cover.



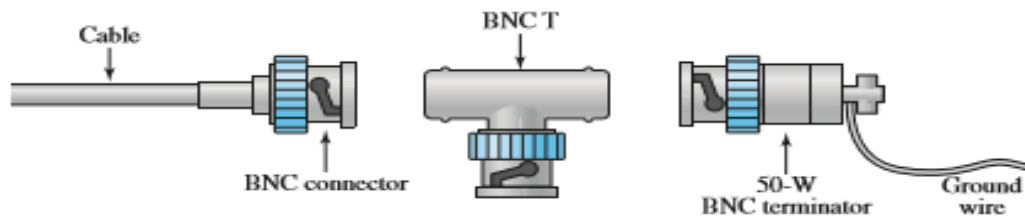
- Coaxial cables are categorized by their Radio Government (RG) ratings.
- Each RG number denotes a unique set of physical specifications, including the wire gauge of the inner conductor, the thickness and type of the inner insulator, the construction of the shield, and the size and type of the outer casing.

- Each cable defined by an RG rating is adapted for a specialized function.

<i>Category</i>	<i>Impedance</i>	<i>Use</i>
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

Coaxial cable connectors:

- To connect coaxial cable to devices, we need coaxial connectors. The most common type of connector used today is the Bayonet Neill-Concelman (BNC) connector.
- Three popular types of these connectors: the BNC connector, the BNC T connector, and the BNC terminator.



- The BNC connector is used to connect the end of the cable to a device, such as a TV set.
- The BNC T connector is used in Ethernet networks to branch out to a connection to a computer or other device.
- The BNC terminator is used at the end of the cable to prevent the reflection of the signal.

❖ Advantages:

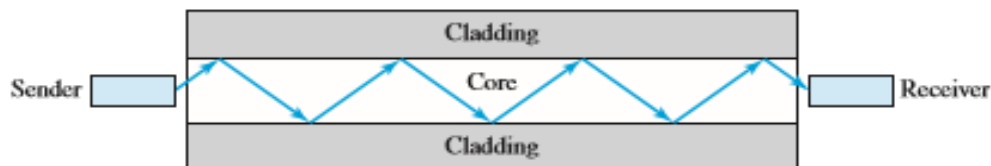
- It is reliable and can carry for higher data rates.
- High Bandwidth.
- Cross talk is less.
- It supports long distance.

❖ Disadvantages:

- More expensive.
- Less flexible.
- More difficult to install.

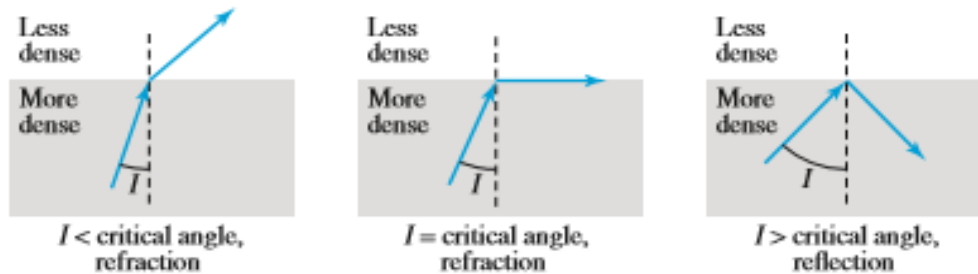
Fiber-optic cable:

- A fiber-optic cable is made of glass or plastic and transmits signals in the form of light.
- This cable can transmit information voice, video as well as data in the form light.
- An optical fiber consists of core, cladding and protecting cover.

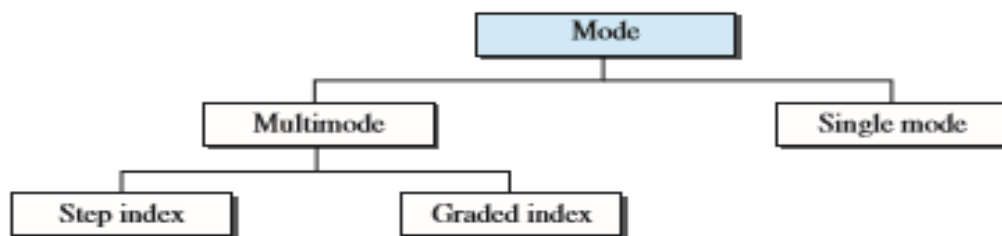


- Light travels in a straight line as long as it is moving through a single uniform substance. If a ray of light traveling through one substance suddenly enters another substance (of a different density), the ray changes direction.

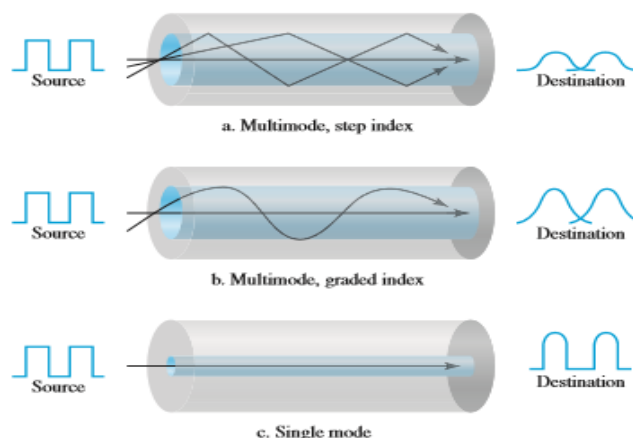
- **Angle of incidence I** (the angle the ray makes with the line perpendicular to the interface between the two substances) is less than the critical angle; the ray refracts and moves closer to the surface.
- If the angle of incidence is equal to the critical angle, the light bends along the interface.
- If the angle is greater than the critical angle, the ray reflects (makes a turn) and travels again in the same medium.



- **Propagation Modes**
- Current technology supports two modes (multimode and single mode) for propagating light along optical channels, each requiring fiber with different physical characteristics.
- Multimode can be implemented in two forms: step-index or graded-index.

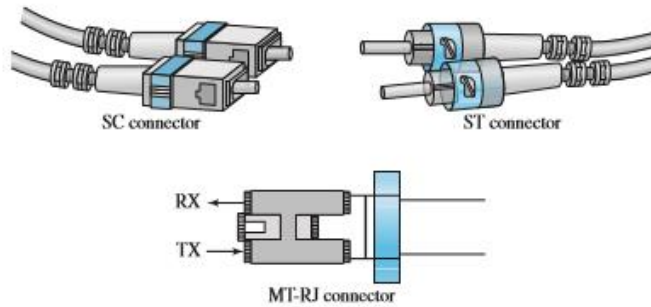


- Multimode is so named because multiple beams from a light source move through the core in different paths. How these beams move within the cable depends on the structure of the core.
- In **multimode step-index fiber**, the density of the core remains constant from the center to the edges. A beam of light moves through this constant density in a straight line until it reaches the interface of the core and the cladding.
- A second type of fiber, called **multimode graded-index fiber**, decreases this distortion of the signal through the cable.
- **Single-Mode:** Single-mode uses step-index fiber and a highly focused source of light that limits beams to a small range of angles, all close to the horizontal.



Fiber-optic cable connectors:

- There are three types of connectors for fiber-optic cables.
- The **subscriber channel (SC)** connector is used for cable TV. It uses a push/pull locking system.



- The **straight-tip (ST) connector** is used for connecting cable to networking devices. It uses a bayonet locking system and is more reliable than SC.
- **MT-RJ** is a connector that is the same size as RJ45.

❖ Advantages:

- Very high bandwidth.
- Very long distance.
- High security.
- Small in size.
- More flexible and strong.

❖ Disadvantages:

- Very expensive.
- Adding node is difficult.
- Special equipment is needed for installation.

Unguided media (wireless):

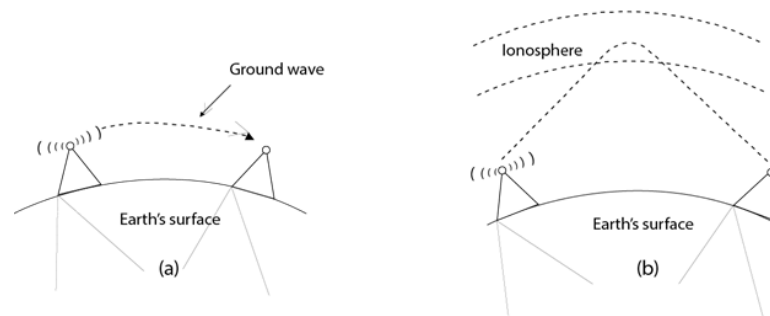
- Unguided medium transport electromagnetic waves without using a physical medium.
- This type of communication is often referred to as wireless communication.
- Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.
- Unguided transmission is broadly classified into three categories:
 - Radio waves.
 - Microwaves.
 - Infrared.

Radio waves:

- Radio waves are the electromagnetic waves that are transmitted in all the directions of free space.
- Radio waves use omnidirectional antennas that send out signals in all directions.



- The range in frequencies of radio waves is from 3 KHz to 1 GHz.
- In the case of radio waves, the sending and receiving antenna are not aligned, i.e., the wave sent by the sending antenna can be received by any receiving antenna.
- An example of the radio wave is **FM radio**.

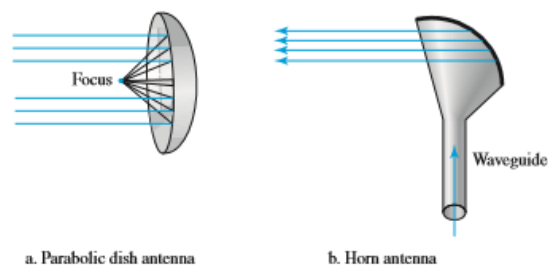


Applications of radio waves:

- The omnidirectional characteristics of radio waves make them useful for multicasting, in which there is one sender but many receivers.
- AM and FM radio, television, maritime radio, cordless phones, and paging are examples of multicasting.

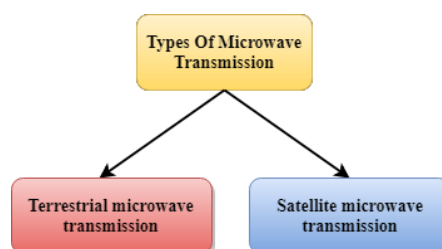
Microwaves:

- Electromagnetic waves having frequencies between 1 and 300 GHz are called microwaves.
- Microwaves are unidirectional.
- The following describes some characteristics of microwave propagation:
 - ☐ Microwave propagation is line-of-sight.
 - ☐ Very high-frequency microwaves cannot penetrate walls.
 - ☐ The microwave band is relatively wide, almost 299 GHz.
 - ☐ Use of certain portions of the band requires permission from authorities.
- Microwaves need unidirectional antennas that send out signals in one direction.
- Two types of antennas are used for microwave communications: the parabolic dish and the horn.



Microwaves are of two types:

- Terrestrial microwave
- Satellite microwave communication.



Terrestrial Microwave Transmission

- Terrestrial Microwave transmission is a technology that transmits the focused beam of a radio signal from one ground-based microwave transmission antenna to another.
- It works on the line of sight transmission.

Satellite Microwave Communication

- A satellite is a physical object that revolves around the earth at a known height.
- Satellite communication is more reliable nowadays as it offers more flexibility than cable and fiber optic systems.
- We can communicate with any point on the globe by using satellite communication.

Applications of Microwaves:

- Microwaves due to their unidirectional properties, are very useful when unicast (one-to-one) communication is needed between the sender and the receiver.
- They are used in cellular phones, satellite networks, and wireless LANs.

Infrared:

- Infrared waves, with frequencies from 300 GHz to 400 THz (wavelengths from 1 mm to 770 nm), can be used for short-range communication.
- Infrared waves, having high frequencies, cannot penetrate walls.

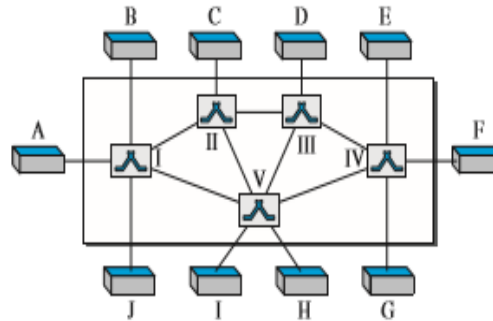
Applications of infrared:

- The infrared band, almost 400 THz, has an excellent potential for data transmission.
- Such a wide bandwidth can be used to transmit digital data with a very high data rate. The Infrared Data Association (IrDA) has established standards for using these signals for communication between devices such as keyboards, mice, PCs, and printers.
- For example, some manufacturers provide a special port called the IrDA port that allows a wireless keyboard to communicate with a PC.
- The standard originally defined a data rate of 75 kbps for a distance up to 8 m.
- The recent standard defines a data rate of 4 Mbps.

❖ Switching :

- A network is a set of connected devices. Whenever we have multiple devices, we have the problem of how to connect them to make one-to-one communication possible.
- One solution is to make a point-to-point connection between each pair of devices (a mesh topology) or between a central device and every other device (a star topology).
- These methods, however, are impractical and wasteful when applied to very large networks.
- The number and length of the links require too much infrastructure to be cost-efficient, and the majority of those links would be idle most of the time.
- Other topologies employing multipoint connections, such as a bus, are ruled out because the distances between devices and the total number of devices increase beyond the capacities of the media and equipment.
- A better solution is switching.
- A switched network consists of a series of interlinked nodes, called switches. Switches are devices capable of creating temporary connections between two or more devices linked to the switch.

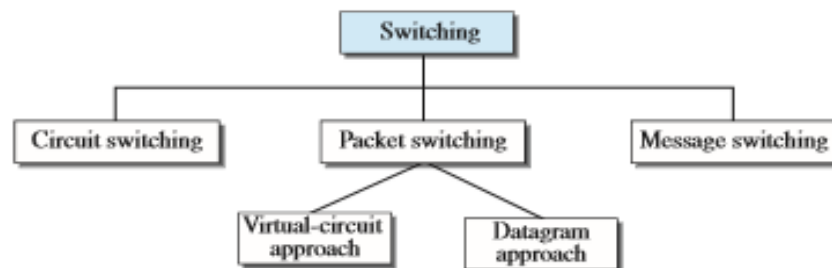
- In a switched network, some of these nodes are connected to the end systems (computers or telephones, for example). Others are used only for routing.



- The end systems (communicating devices) are labeled A, B, C, D, and so on, and the switches are labeled I, II, III, IV, and V. Each switch is connected to multiple links.

Three Methods of Switching:

- Traditionally, three methods of switching have been discussed: circuit switching, packet switching, and message switching.
- The first two are commonly used today.
- The third has been phased out in general communications but still has networking applications.
- Packet switching can further be divided into two subcategories
 - virtual circuit approach
 - Datagram approach



Circuit switching Network:

- A circuit-switched network consists of a set of switches connected by physical links.
- A connection between two stations is a dedicated path made of one or more links.
- However, each connection uses only one dedicated channel on each link.
- Each link is normally divided into n channels by using FDM or TDM.
- ☐ Circuit switching takes place at the physical layer.
- ☐ Before starting communication, the stations must make a reservation for the resources to be used during the communication. These resources, such as channels (bandwidth in FDM and time slots in TDM).
- ☐ Data transferred between the two stations are not packetized (physical layer transfer of the signal).
- ☐ There is no addressing involved during data transfer. The switches route the data based on their occupied band (FDM) or time slot (TDM).
- ❖ **Three Phases :**
 - The actual communication in a circuit-switched network requires three phases: “connection setup, data transfer, and connection teardown”.

- **Setup Phase:** Before the two parties (or multiple parties in a conference call) can communicate, a dedicated circuit (combination of channels in links) needs to be established. The end systems are normally connected through dedicated lines to the switches, so connection setup means creating dedicated channels between the switches.
- **Data-Transfer Phase:** After the establishment of the dedicated circuit (channels), the two parties can transfer data.
- **Teardown Phase:** When one of the parties needs to disconnect, a signal is sent to each switch to release the resources.
- Let us use a circuit-switched network to connect eight telephones in a small area.

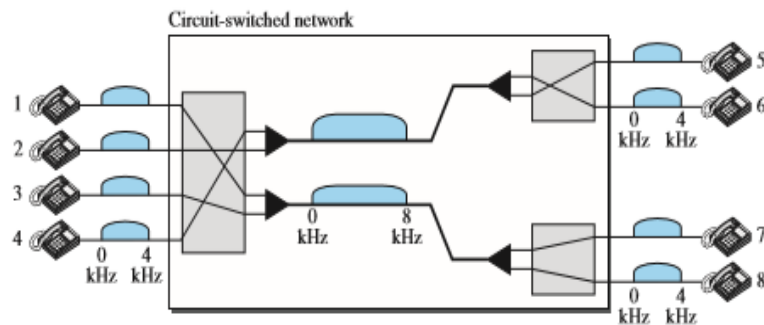


Fig: Circuit switching Network

- Communication is through 4-kHz voice channels.
- We assume that each link uses FDM to connect a maximum of two voice channels.
- The bandwidth of each link is then 8 kHz. Telephone 1 is connected to telephone 7; 2 to 5; 3 to 8; and 4 to 6.
- Of course the situation may change when new connections are made.
- The switch controls the connections.

➤ Efficiency

- It can be argued that circuit-switched networks are not as efficient as the other two types of networks because resources are allocated during the entire duration of the connection. These resources are unavailable to other connections.

➤ Delay

- Although a circuit-switched network normally has low efficiency, the delay in this type of network is minimal. During data transfer the data are not delayed at each switch; the resources are allocated for the duration of the connection.

Packet switching Network:

- In data communications, we need to send messages from one end system to another.
- If the message is going to pass through a packet-switched network, it needs to be divided into packets of fixed or variable size.
- The size of the packet is determined by the network and the governing protocol.
- In packet switching, there is no resource allocation for a packet. This means that there is no reserved bandwidth on the links, and there is no scheduled processing time for each packet.
- Resources are allocated on demand. The allocation is done on a first come, first-served basis.
- When a switch receives a packet, no matter what the source or destination is, the packet must wait if there are other packets being processed.

- We can have two types of packet-switched networks:
 - Datagram networks
 - Virtual circuit networks.
- ❖ **Datagram Networks:**
 - In a datagram network, each packet is treated independently of all others.
 - Even if a packet is part of a multi packet transmission, the network treats it as though it existed alone.
 - Packets in this approach are referred to as datagram.
 - Datagram switching is normally done at the network layer.

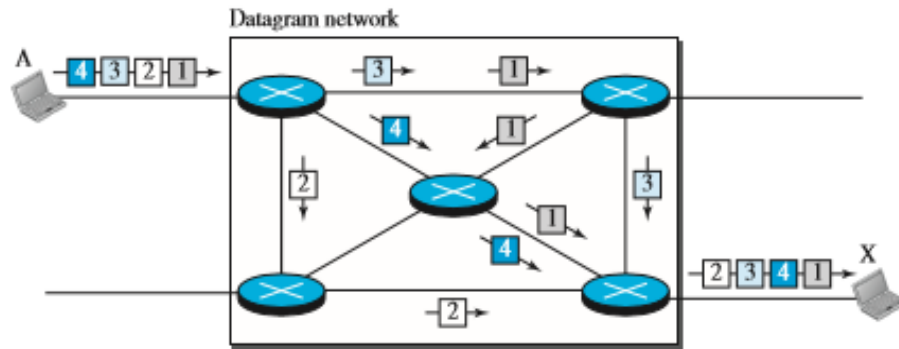


Fig: Datagram Networks

- The datagram networks are sometimes referred to as connectionless networks.
- **Routing Table:** If there are no setup or teardown phases, how are the packets routed to their destinations in a datagram network? In this type of network, each switch (or packet switch) has a routing table which is based on the destination address. The routing tables are dynamic and are updated periodically.
- **Destination Address:** Every packet in a datagram network carries a header that contains, among other information, the destination address of the packet. When the switch receives the packet, this destination address is examined; the routing table is consulted to find the corresponding port through which the packet should be forwarded.
- **Efficiency:** The efficiency of a datagram network is better than that of a circuit-switched network; resources are allocated only when there are packets to be transferred.
- **Delay:** There may be greater delay in a datagram network than in a virtual-circuit network. Although there are no setup and teardown phases, each packet may experience a wait at a switch before it is forwarded.
- ❖ **Virtual-Circuit Packet Networks:**
 - A virtual-circuit network is a cross between a circuit-switched network and a datagram network.
 - It has some characteristics of both.
 - As in a circuit-switched network, there are setup and teardown phases in addition to the data transfer phase.
 - Resources can be allocated during the setup phase, as in a circuit-switched network, or on demand, as in a datagram network.
 - As in a datagram network, data are packetized and each packet carries an address in the header. However, the address in the header has local jurisdiction.
 - As in a circuit-switched network, all packets follow the same path established during the connection.

- A virtual-circuit network is normally implemented in the data-link layer, while a circuit-switched network is implemented in the physical layer and a datagram network in the network layer.
- The network has switches that allow traffic from sources to destinations.
- A source or destination can be a computer, packet switch, bridge, or any other device that connects other networks.

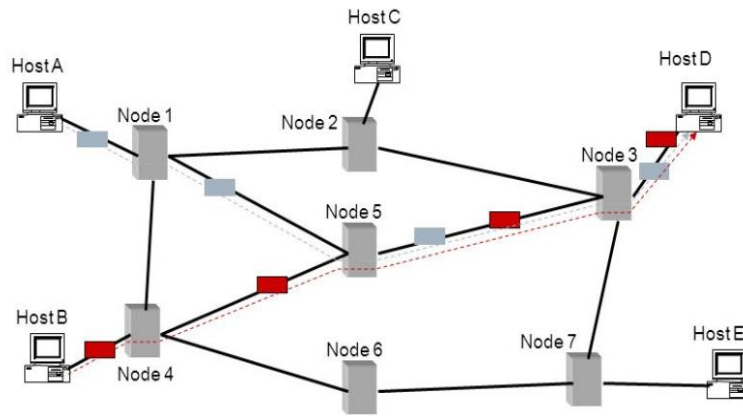
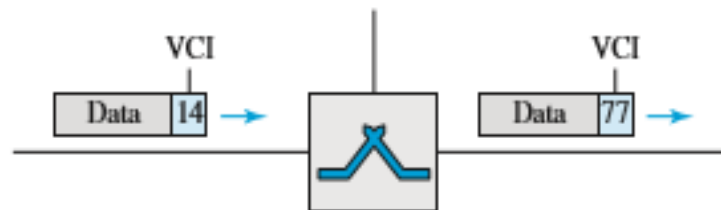


Fig: Virtual circuit packet switching

- **Addressing:** In a virtual-circuit network, two types of addressing are involved: global and local (virtual-circuit identifier).
- **Global Addressing:** A source or a destination needs to have a global address—an address that can be unique in the scope of the network or internationally if the network is part of an international network.
- **Virtual-Circuit Identifier:** The identifier that is actually used for data transfer is called the virtual-circuit identifier (VCI) or the label. A VCI, unlike a global address, is a small number that has only switch scope; it is used by a frame between two switches.



❖ Three Phases :

- As in a circuit-switched network, a source and destination need to go through three phases in a virtual-circuit network: setup, data transfer, and teardown.
- **Setup Phase:** In the setup phase, a switch creates an entry for a virtual circuit. For example, suppose source A needs to create a virtual circuit to B. Two steps are required: the **setup request** and the **acknowledgment**.
- A setup request frame is sent from the source to the destination.
- A special frame, called the acknowledgment frame, completes the entries in the switching tables. The destination sends an acknowledgment to source.
- **Data-Transfer Phase:** To transfer a frame from a source to its destination, all switches need to have a table entry for this virtual circuit. The table, in its simplest form, has four columns.
- **Teardown Phase:** When one of the parties needs to disconnect, a signal is sent to each switch to release the resources.

➤ **Efficiency:**

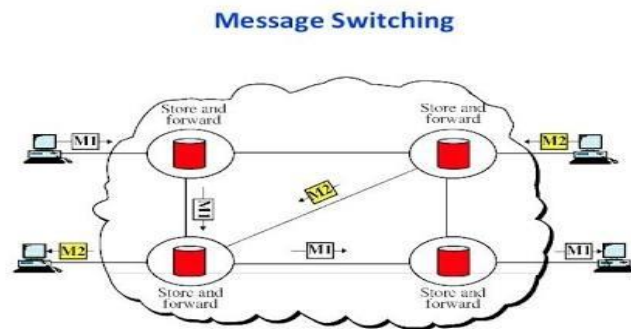
- Resource reservation in a virtual-circuit network can be made during the setup or can be on demand during the data-transfer phase.
- In the first case, the delay for each packet is the same; in the second case, each packet may encounter different delays.
- There is one big advantage in a virtual-circuit network even if resource allocation is on demand.

➤ **Delay in Virtual-Circuit Networks**

- In a virtual-circuit network, there is a one-time delay for setup and a one-time delay for teardown.
- If resources are allocated during the setup phase, there is no wait time for individual packets.
- The delay for a packet traveling through two switches in a virtual-circuit network.

Message switching Networks:

- Message switching is a network switching technique in which data is routed in its entirety from the source node to the destination node, one hop at a time.
- During message routing, every intermediate switch in the network stores the whole message until the appropriate route is free and then sends it along that route.



❖ **STRUCTURE OF A SWITCH**

- We use switches in circuit-switched and packet-switched networks. In this section, we discuss the structures of the switches used in each type of network.

➤ **Structure of Circuit Switches**

➤ **Structure of Packet Switches**

❖ **Structure of Circuit Switches**

- Circuit switching today can use either of two technologies:

- ☐ The space-division switch
- ☐ The time-division switch.

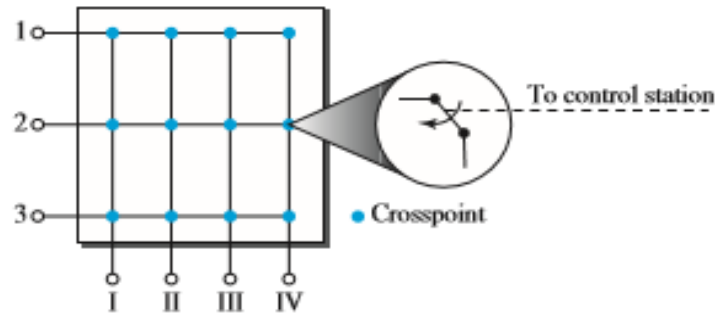
➤ **Space-Division Switch:** In space-division switching, the paths in the circuit are separated from one another spatially.

➤ This technology was originally designed for use in analog networks but is used currently in both analog and digital networks.

- ☐ Crossbar Switch
- ☐ Multistage Switch

➤ **Crossbar Switch**

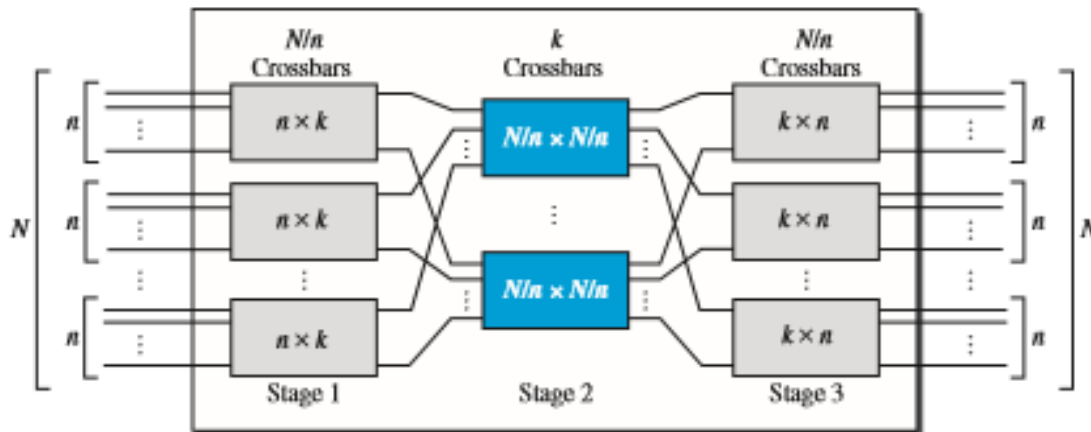
- A crossbar switch connects n inputs to m outputs in a grid, using electronic micro switches (transistors) at each cross point.
- The major limitation of this design is the number of cross points required.
- To connect n inputs to m outputs using a crossbar switch requires $n \times m$ cross points.



- For example, to connect 1000 inputs to 1000 outputs requires a switch with 1,000,000 cross points.
- A crossbar switch [?] with this number of cross points is impractical.
- Such a switch is also inefficient because statistics show that, in practice, fewer than 25 percent of the cross points are in use at any given time. The rest are idle.

➤ Multistage Switch

- The solution to the limitations of the crossbar switch is the multistage switch, which combines crossbar switches in several (normally three) stages.
- In a single crossbar switch, only one row or column (one path) is active for any connection. So we need $N \times N$ cross points.
- If we can allow multiple paths inside the switch, we can decrease the number of cross points.
- Each cross point in the middle stage can be accessed by multiple cross points in the first or third stage.



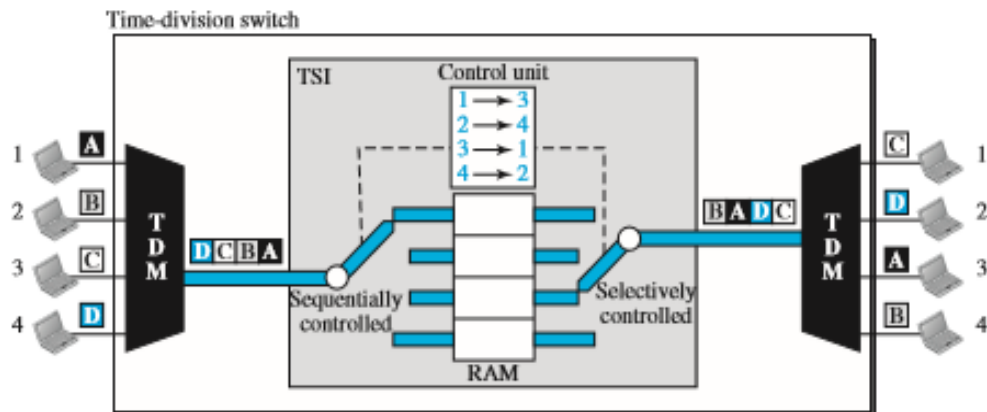
- To design a three-stage switch, we follow these steps:
- We divide the N input lines into groups, each of n lines. For each group, we use one crossbar of size $n \times k$, where k is the number of crossbars in the middle stage. In other words, the first stage has N/n crossbars of $n \times k$ cross points.
- We use k crossbars, each of size $(N/n) \times (N/n)$ in the middle stage.
- We use N/n crossbars, each of size $k \times n$ at the third stage. We can calculate the total number of cross points as follows:

$$\frac{N}{n} (n \times k) + k \left(\frac{N}{n} \times \frac{N}{n} \right) + \frac{N}{n} (k \times n) = 2kN + k \left(\frac{N}{n} \right)^2$$

➤ Time-Division Switch:

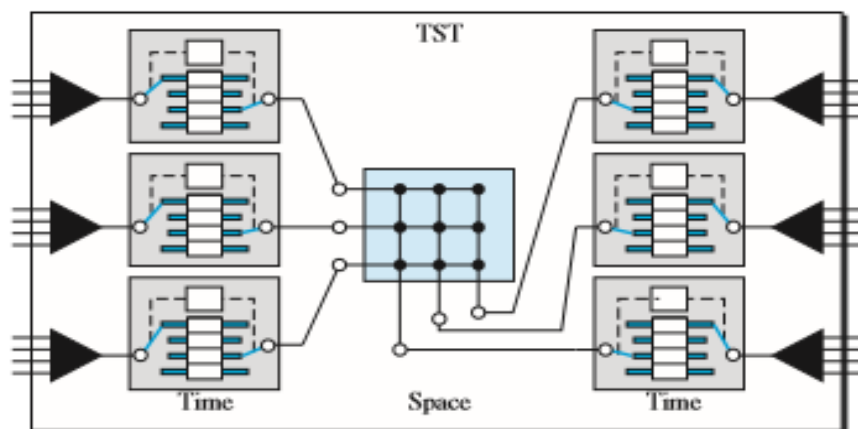
- Time-division switching uses time-division multiplexing (TDM) inside a switch. The most popular technology is called the time-slot interchange (TSI).

- Time-Slot Interchange a system connecting four input lines to four output lines. Imagine that each input line wants to send data to an output line according to the following pattern: (1 → 3), (2 → 4), (3 → 1), and (4 → 2), in which the arrow means “to.”



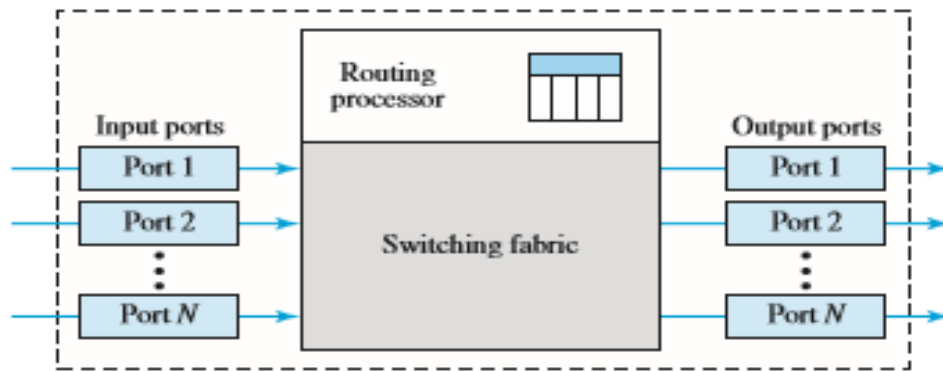
➤ Time- and Space-Division Switch Combinations:

- When we compare space-division and time-division switching, some interesting facts emerge.
- The advantage of space-division switching is that it is instantaneous.
- Its disadvantage is the number of cross points required to make space-division switching acceptable in terms of blocking.
- The advantage of time-division switching is that it needs no cross points.
- Its disadvantage, in the case of TSI, is that processing each connection creates delays. Each time slot must be stored by the RAM, then retrieved and passed on.
- In a third option, we combine space-division and time-division technologies to take advantage of the best of both.
- Combining the two results in switches that are optimized both physically (the number of cross points) and temporally (the amount of delay).
- Multistage switches of this sort can be designed as time-space-time (TST) switches.

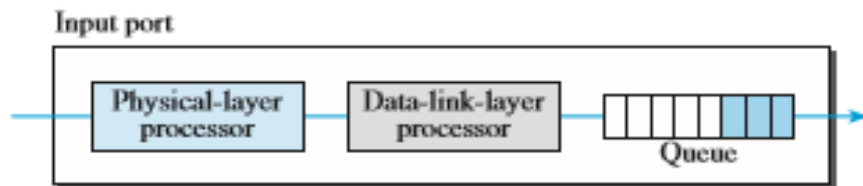


❖ Structure of Packet Switches

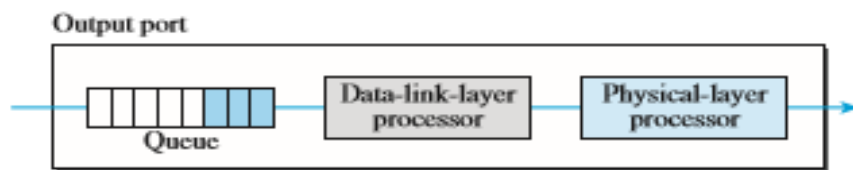
- A switch used in a packet-switched network has a different structure from a switch used in a circuit-switched network.
- We can say that a packet switch has four components: input ports, output ports, the routing processor, and the switching fabric.



- **Input Ports:** An input port performs the physical and data-link functions of the packet switch. The bits are constructed from the received signal.



- **Output Ports:** The output port performs the same functions as the input port, but in the reverse order. First the outgoing packets are queued, then the packet is encapsulated in a frame, and finally the physical-layer functions are applied to the frame to create the signal to be sent on the line.



➤ **Routing Processor**

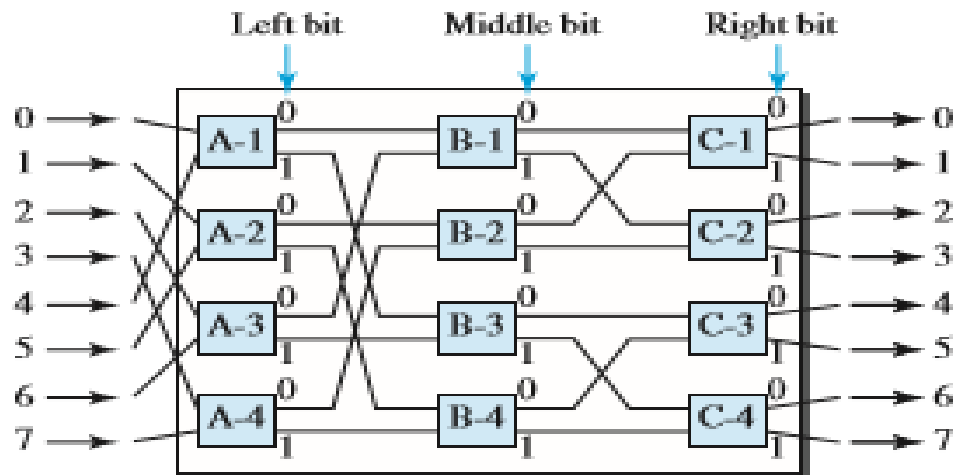
- The routing processor performs the functions of the network layer.
- The destination address is used to find the address of the next hop and, at the same time, the output port number from which the packet is sent out.
- This activity is sometimes referred to as table lookup because the routing processor searches the routing table.
- In the newer packet switches, this function of the routing processor is being moved to the input ports to facilitate and expedite the process.

➤ **Switching Fabrics**

- The most difficult task in a packet switch is to move the packet from the input queue to the output queue.
- The speed with which this is done affects the size of the input/output queue and the overall delay in packet delivery.
- In the past, when a packet switch was actually a dedicated computer, the memory of the computer or a bus was used as the switching fabric.
- The input port stored the packet in memory; the output port retrieved the packet from memory.
- Today, packet switches are specialized mechanisms that use a variety of switching fabrics.
- The simplest type of switching fabric is the crossbar switch, discussed in the previous section.

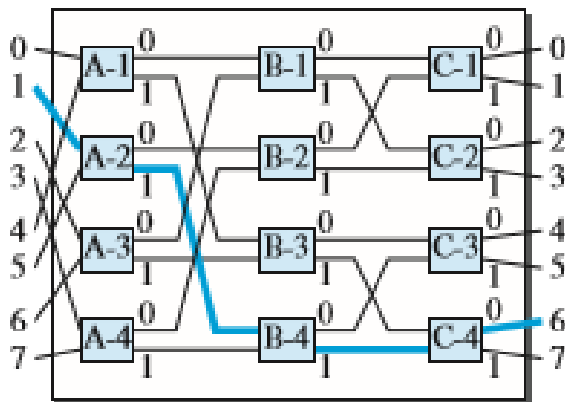
➤ Banyan Switch

- A more realistic approach than the crossbar switch is the banyan switch (named after the banyan tree).
- A banyan switch is a multistage switch with micro switches at each stage that route the packets based on the output port represented as a binary string.
- For n inputs and n outputs, we have $\log_2 n$ stages with $n/2$ micro switches at each stage.
- The first stage routes the packet based on the high-order bit of the binary string.
- The second stage routes the packet based on the second high-order bit, and so on.
- A banyan switch with eight inputs and eight outputs. The number of stages is $\log_2(8) = 3$.

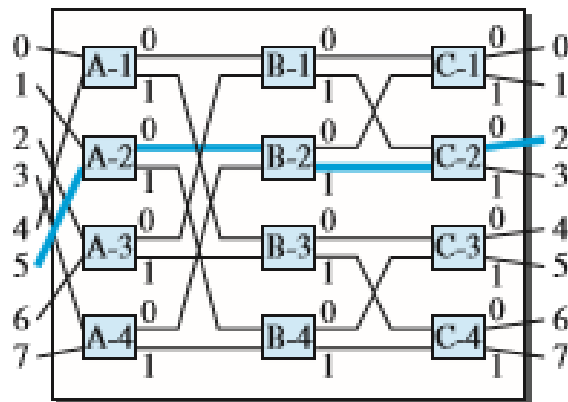


• Examples of routing in a banyan switch

- In part a, a packet has arrived at input port 1 and must go to output port 6 (110 in binary). The first micro switch (A-2) routes the packet based on the first bit (1), the second micro switch (B-4) routes the packet based on the second bit (1), and the third micro switch (C-4) routes the packet based on the third bit (0).



a. Input 1 sending a cell to output 6 (110)



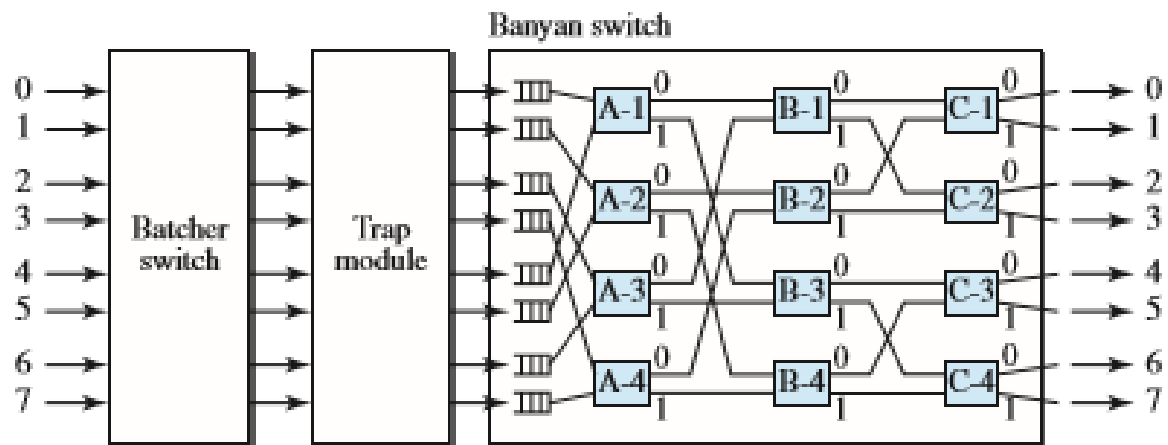
b. Input 5 sending a cell to output 2 (010)

- In part b, a packet has arrived at input port 5 and must go to output port 2 (010 in binary). The first micro switch (A-2) routes the packet based on the first bit (0), the second micro switch (B-2) routes the packet based on the second bit (1), and the third micro switch (C-2) routes the packet based on the third bit (0).

➤ Batcher-Banyan Switch

- The problem with the banyan switch is the possibility of internal collision even when two packets are not heading for the same output port.
- We can solve this problem by sorting the arriving packets based on their destination port.

- K. E. Batcher designed a switch that comes before the banyan switch and sorts the incoming packets according to their final destinations.
- The combination is called the Batcher-banyan switch.
- The sorting switch uses hardware merging techniques.
- Normally, another hardware module called a trap is added between the Batcher switch and the banyan switch.



- The trap module prevents duplicate packets (the packets with the same output destination) from passing to the banyan switch simultaneously.
- Only one packet for each destination is allowed at each tick; if there is more than one, they wait for the next tick.

